sbuck-5v3a — Design Document

PCB-0008-B

hwde v1 pipeline

generated 2026-09-25 18:10:53

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1 Board and Run Metadata

Field	Value
board	sbuck-5v3a
part number	PCB-0008-B
workspace	`boards/PCB-0008-B_sbuck-5v3a`
phase	P10
gate status	all 5 recorded gates pass
state created	2026-08-08T22:12:30
state updated	2026-08-09T06:59:55

2 Overview

Brief - synchronous buck converter board (sbuck-5v3a)

Design a synchronous buck converter board.

SPECIFICATION

- Input: 7-18V DC, nominal 12V, via 2-pin 5.08mm screw terminal
- Output: 5.0V +/-2%, 3A continuous, via 2-pin 5.08mm screw terminal
- Load step: 0 to 3A, recovery within 100us, <200mV excursion
- Output ripple: <50mV pk-pk at full load

- Efficiency: >88% at 12V in, 3A out
- Ambient: up to 50C, no forced airflow
- Protection: input reverse polarity, output overcurrent, thermal shutdown
- Board: <=50x40mm, 4x M3 mounting holes, 2 or 4 layer (your call, justify it)

PART SELECTION

- Prefer an integrated synchronous buck IC over a controller + discrete FETs at this power level. Candidate families to evaluate: TI TPS54xxx/TPS56xxx, MPS MP23xx/MP15xx, Diodes AP63xxx, Silergy SY82xx. Do not trust remembered ratings for any of these - pull the datasheet and confirm Vin range, Iout, and switching frequency before committing.
- Verify LCSC stock and basic/extended tier for every part before finalising. A part that is out of stock is a failed design.
- Inductor: compute ripple current dI = (Vin - Vout) * D / (L * fsw), where D = Vout/Vin. Target 20-40% of full load current. Size saturation current above Iout + dI/2 with margin, at temperature, not at 25C. Check DCR against your efficiency target.
- Input capacitance: ceramic close-in for the switching loop plus bulk for the cable inductance. Input cap RMS current is roughly Iout * sqrt(D * (1-D)) - check the ceramic's ripple rating and derate MLCC capacitance for DC bias, which can cost more than half the nameplate value at rated voltage.
- Output capacitance: ripple is roughly dI/(8 * fsw * Cout) + dI * ESR. Check the load-step requirement too, which usually sets Cout above what ripple alone needs.
- Every MLCC needs voltage derating; X7R or X5R only, no Y5V.

SCHEMATIC

- Feedback divider sized for the IC's internal reference; keep divider impedance low enough that FB bias current does not shift the setpoint.
- Include the bootstrap capacitor, enable/UVLO divider if the IC supports it, and a soft-start component if it is not internal.
- Reverse polarity: a P-channel FET in the supply path is more efficient than a series diode at 3A. Confirm Vgs rating against maximum input voltage.
- Add an optional RC snubber footprint across the low-side switch, DNP by default.
- Provide test points for VIN, SW, VOUT, FB, EN, GND, plus a low-inductance scope ground point near the output.

LAYOUT

Placement first, routing second. Placement decides this design.

- The high di/dt loop is input capacitor -> high-side switch -> low-side switch -> back to the capacitor's ground. Minimise its enclosed area above everything else. Put the smallest input ceramic directly across the IC's VIN and PGND pins on the same layer, no vias in that loop.
- The switch node is the primary radiator. Make its copper large enough to carry current and conduct heat, and no larger. Do not pour it as a plane.

- Keep the feedback trace short, away from SW and the inductor body, and tap it from the output capacitor terminal rather than the inductor pin so it senses the regulated node.
- Bootstrap capacitor directly at its pins.
- Single low-impedance ground reference under the converter. Avoid routing signal returns through the power ground path. If 4 layer, put an uninterrupted ground plane on the layer directly under the switching components and do not cut it.
- Size copper by current using IPC-2152, accounting for the 10-20C rise you can accept. 1oz outer copper is 35um; check whether inner layers are 0.5oz in the stackup you pick.
- Place a via array under the IC's thermal pad and stitch the ground planes together around the loop.

THERMAL

- Estimate dissipation at 12V/3A from the IC's efficiency curve, then check junction temperature against RthJA for the copper area you actually have, at 50C ambient.
- Inductor DCR loss is I^2 * DCR; confirm the part's own temperature rise.
- If the numbers do not close, increase copper area or change parts. Do not assume the exposed pad alone is sufficient.

EMI / DFM

- Keep the switching loop and SW node away from the board edge and connectors.
- No unnecessary ground plane splits or slots under the converter.
- Silkscreen legible, none on pads, polarity and pin 1 marked on every polarised part.
- Respect JLCPCB's minimum trace/space, annular ring, and hole sizes for the process class you select. Confirm every footprint's rotation against JLCPCB's CPL convention, not the KiCad default.

BEFORE DECLARING DONE

1. DRC and ERC clean, zero unrouted nets. 2. Recompute inductor peak current, input cap RMS current, output ripple, and junction temperature from the parts you actually chose, and state the numbers. 3. Confirm each part is in stock at LCSC with its part number recorded. 4. Report the input loop enclosed area, switch node copper area, feedback trace length, and thermal via count.

DECISION POLICY

Make every remaining decision yourself. Do not ask clarifying questions. Where this brief is silent, take the conventional, conservative option. Record each non-obvious choice in DECISIONS.md with one line of reasoning.

Deliver: schematic, layout, gerbers, BOM, CPL, DECISIONS.md, and the computed numbers above.

3 Requirements

requirements.md - sbuck-5v3a

Source: `brief/brief.md` (sole input; no attached datasheets, drawings or reference schematics were supplied).

Note on process: the brief carries an explicit DECISION POLICY - "Make every remaining decision yourself. Do not ask clarifying questions. Where this brief is silent, take the conventional, conservative option." Section 9 below therefore lists the unknowns as questions WITH a recommended default each, so the orchestrator can rule on them as delegate and record them in DECISIONS.md. Nothing in section 9 is blocking on the user.

Legend: `STATED` = written in the brief. `DERIVED` = arithmetic from stated numbers. `GUESS` / `ASSUMED` = not in the brief, low risk, recorded so it can be overridden. `HARD` = binds permanently at P5 board_init.

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1. Function

A standalone, open-frame step-down DC-DC power module: it takes an unregulated 7-18 V DC input (12 V nominal) on a 2-pin screw terminal and produces a regulated 5.0 V +/-2% output at up to 3 A continuous (15 W) on a second 2-pin screw terminal, using a synchronous buck topology built around an integrated buck IC (internal high-side and low-side switches) rather than a controller plus discrete FETs. The board protects itself and its source against reverse-polarity connection of the input, and protects itself against output overcurrent and over-temperature. There is no MCU, no digital interface, no telemetry, and no sequencing or remote control: the board regulates whenever a valid input is applied. Scope is the converter only.

2. Interfaces

#	Interface	Dir	Electrical	Connector
I1	DC input	in	7-18 V DC, 12 V nominal. Worst-case input current ~2.6 A at 7 V in / 3 A out (DERIVED). Reverse-polarity tolerant.	2-pin 5.08 mm pitch screw terminal (STATED). Exact part/entry direction: Q9
I2	DC output	out	5.0 V +/-2% (4.90-5.10 V), 3 A continuous.	2-pin 5.08 mm pitch screw terminal (STATED)
I3	Test point VIN	probe	post-reverse-polarity-FET input rail	test pad, see Q13
I4	Test point SW	probe	switch node, high dv/dt - short-stub tap only, must not extend SW copper	test pad, see Q13
I5	Test point VOUT	probe	regulated output, tapped at the output cap terminal	test pad, see Q13
I6	Test point FB	probe	feedback node - high impedance, probe loading must not shift setpoint	test pad, see Q13
I7	Test point EN	probe	enable / UVLO node	test pad, see Q13
I8	Test point GND	probe	general ground reference	test pad, see Q13
I9	Scope ground	probe	dedicated LOW-INDUCTANCE ground point adjacent to the VOUT test point, sized for a scope spring/bayonet ground rather than a ground lead (STATED)	see Q13

Non-connector interface items stated in the brief:

- RC snubber footprint across the low-side switch (i.e. SW to PGND), populated DNP by default. Footprint must exist and be routable with low inductance even though unstuffed.
- No indicator LED, no PG output, no external enable input, no adjustable-output feature, and no remote sense are stated. See Q11 (LED) and Q14 (EN control) for the defaults.

3. Power

Input (STATED)

Param	Value
Range	7.0 - 18.0 V DC
Nominal	12.0 V
Absolute max at connector	18 V (no surge/load-dump spec given - see Q6)
Source impedance / cabling	not stated; brief requires bulk input capacitance for "the cable inductance", so a non-trivial cable run is implied

Output (STATED)

Param	Value
Voltage	5.0 V, +/-2% = 4.90-5.10 V (window scope: see Q1)
Current	3.0 A continuous
Power	15.0 W (DERIVED)
Ripple	< 50 mV pk-pk at full load (measurement bandwidth: see Q2)
Load step	0 -> 3 A and 3 A -> 0: excursion < 200 mV, recovery within 100 us (slew rate of the step: see Q3)
Efficiency	> 88% at 12 V in, 3 A out (measurement points: see Q4)

Derived operating points (arithmetic from the stated spec; not guesses)

Param	At Vin = 7 V	At Vin = 12 V	At Vin = 18 V
Duty D = Vout/Vin (ideal)	0.714	0.417	0.278
Input current at 3 A out, 88% eff	2.44 A	1.42 A	0.95 A
Input cap RMS current, Iout*sqrt(D(1-D))	1.36 A	1.48 A	1.34 A

- Worst-case input capacitor RMS current over the whole input range is 1.50 A, at D = 0.5, i.e. Vin = 10 V - which is INSIDE the operating range. Size the input ceramics for 1.5 A RMS, not for the 1.48 A at the 12 V nominal point.
- Board dissipation at the efficiency floor: Pin = 15.0/0.88 = 17.05 W, so Ploss = 2.05 W total at 12 V/3 A. This is the thermal budget the 50 C ambient case must close against.
- Reverse-polarity P-FET conducts the full input current (2.44 A worst case at 7 V) and its I^2*Rds(on) loss counts against the 88% number if efficiency is measured terminal to terminal (Q4).

Rails Single output rail. No auxiliary rails, no bias rail, no battery, no charging, no energy storage beyond the converter's own input/output capacitance.

Protection (all STATED as requirements)

P	Requirement	Notes
P1	Input reverse polarity	Brief directs a P-channel FET in the supply path (not a series diode) and requires its Vgs rating be confirmed against max Vin. Vgs clamp/divider needed if |Vgs| rating < 18 V.
P2	Output overcurrent	Mechanism not specified - see Q7
P3	Thermal shutdown	Not specified whether IC-internal is sufficient - see Q8

4. Environment

Param	Value	Status
Max operating ambient	50 C	STATED
Airflow	none - natural convection only	STATED
Min operating ambient	not stated	Q15, default 0 C
Storage range	not stated	ASSUMED: -40 to +85 C, standard commercial part ratings
Enclosure	not stated	Q16 - drives whether 50 C is ambient-air or inside-a-box
Ingress protection	not stated	ASSUMED: none (open frame, indoor, dry)
Vibration / shock	not stated	ASSUMED: none beyond normal handling; no staking or conformal coat
Altitude	not stated	ASSUMED: <= 2000 m, no creepage derating

Note: the thermal requirement is the one that bites. 2.05 W into natural convection at 50 C ambient on a board of at most 50x40 mm, with no heatsink, is the binding constraint of this design, and the brief explicitly forbids assuming the exposed pad alone closes it.

5. Size and mounting

Param	Value	Status
Max outline	50.0 x 40.0 mm	**HARD** (STATED as "<=50x40mm"). Binds at P5 board_init. Smaller is permitted but see Q17 - recommended default is to use the full area for thermal copper.
Layer count	2 or 4, designer's call, must be justified	STATED as a delegated decision - see Q18
Mounting holes	4x M3	**HARD** on count and thread size (STATED). Positions, plating and keepout: Q19
Max component height	not stated	Q20, default 15 mm
Board thickness	not stated	ASSUMED: 1.6 mm FR4, JLCPCB standard
Copper weight	not stated	Q21 - 1 oz outer default, 2 oz escalation path if thermals do not close
Keepouts / connector clearance	not stated	ASSUMED: screw terminals at board edges with wire entry clear of the mounting hardware

6. Quantity and budget

Param	Value	Status
Build quantity	not stated	Q22, default 5 (JLCPCB minimum prototype run)
Target unit cost	not stated	Q23, default: no hard cap; keep BOM under ~$12/board at qty 5 and flag anything that would push a single line item over ~$3
Lifetime / production intent	not stated	ASSUMED: prototype / low-volume bench module, not a mass-production cost-down exercise

Hard commercial constraint that IS stated: every part must be in stock at LCSC with its part number recorded; an out-of-stock part is a failed design. Stock threshold: Q24.

7. Assembly

Param	Value	Status
Assembly method	not stated	Q25, default: JLCPCB PCBA for all SMT
Sides assembled	not stated	Q26, default: single-sided (top) SMT assembly
Through-hole parts	the two 5.08 mm screw terminals are inherently THT	Q25 covers whether to buy JLC's THT service or hand-solder the two connectors

Footprint rotation	must be checked against JLCPCB's CPL convention, NOT the KiCad default	STATED
Process class	must respect JLCPCB minimum trace/space, annular ring and hole size for the selected class	STATED; class choice: Q27
Part tier	LCSC Basic vs Extended not stated	Q28, default: prefer Basic/Preferred for passives, accept Extended where the function demands it (buck IC, inductor, P-FET)

8. Compliance / safety flags

Applicable:

- **High current (>= 3 A).** The output is specified at exactly 3.0 A continuous, which sits on the brief's own high-current threshold, and the input side draws up to ~2.6 A at low line. This is the flag that drives copper sizing (IPC-2152, with the 10-20 C rise the brief permits), connector current rating, and the fusing question. Per role rules this is NOT guessed: see Q5 (is 3 A the true worst case, or is there a peak/inrush above it), Q7 (overcurrent mechanism and whether an input fuse is required) and Q29 (fault behaviour the board must survive: sustained short circuit on the output).
- **Undefined fault energy.** The brief does not state whether the input source is current limited, so the power available into a fault is unbounded as specified: a 4 A input fuse at 18 V still passes ~72 W. Behaviour under a bolted output short, and under a reversed or swapped input, must be defined rather than assumed - Q7, Q29, Q30.
- **Switching EMI.** A hard-switching buck is a deliberate broadband noise source. The brief imposes layout-level EMI requirements (loop area, SW node away from edges/connectors, no plane splits) but names no standard - Q31.

Not applicable (confirmed against the brief):

- No mains voltage anywhere on the board. Maximum potential is 18 V.
- No battery of any chemistry, no charging circuit, no cell balancing.
- No motors, relays, solenoids or other inductive actuator loads driven by this board.
- No node above 30 V under normal operation (18 V max input; switch-node ringing must still be kept below the chosen IC's absolute max, which is a design check, not a safety-agency flag).
- No intentional RF transmission.

Consequence: no agency-certification path (UL/IEC mains safety, battery transport, RF type approval) applies to this board. The safety work is entirely thermal, fault-current and connector-rating engineering.

9. Open questions

Each has a recommended default. The orchestrator answers all of them at once as delegate, per the brief's DECISION POLICY, and each answer becomes a DECISIONS.md line.

1. **Does the +/-2% output window include ripple and load-step transients, or only DC accuracy?** Default: +/-2% is the DC setpoint over line, load and temperature; the 50 mV ripple and the 200 mV load-step excursion are separate, additional allowances. 2. **How is the <50 mV ripple measured?** Default: at the output screw terminal, scope in 20 MHz bandwidth-limited mode with a short spring ground (the standard method) - so high-frequency switching spikes above 20 MHz are excluded from the number. 3. **How fast is the 0-3 A load step?** Default: 2.5 A/us electronic-load slew (a common bench setting), applied at 12 V in. A faster step would be

dominated by output-cap ESL rather than by the control loop. 4. **Is the >88% efficiency measured at the screw terminals (including the reverse-polarity FET and terminal resistance) or across the IC only?** Default: terminal to terminal, i.e. the reverse-polarity FET loss counts against the budget. This is the conservative reading. 5. **Is 3 A the true worst case, or must the board survive brief peaks above it (e.g. a capacitive or lamp load at turn-on)?** Default: 3.0 A is the continuous maximum and the design target; the IC's own current limit is allowed to be the only headroom above it, with the current limit set no lower than 4 A to avoid nuisance foldback. 6. **Any input surge above 18 V (automotive load dump, hot-plug ringing on a long cable)?** Default: no formal surge requirement, but select the IC and input caps for a minimum 28 V rating (>1.5x the 18 V max) so hot-plug ringing on an inductive cable does not destroy the part. 7. **Overcurrent protection mechanism, and is an input fuse required?** Default: rely on the buck IC's internal cycle-by-cycle current limit with hiccup-mode short-circuit protection (auto-recovering), AND fit a slow-blow input fuse or PTC sized ~4 A as a fail-safe against a shorted high-side switch, which the IC cannot protect against. 8. **Is the IC's internal thermal shutdown sufficient, or is an independent thermal cutout required?** Default: internal TSD is sufficient given the low power and the absence of any agency requirement; no separate thermistor or cutout. 9. **Screw terminal part and wire entry direction.** Default: 5.08 mm pitch, 2-pin, horizontal (side) wire entry facing outward from the board edge, rated >= 10 A / 300 V, accepting 12-24 AWG; choose whichever LCSC-stocked family (KF128/KF301-class or equivalent) is in stock in quantity with a verified footprint. 10. **Input and output on opposite board edges, or the same edge?** Default: opposite short edges (input left, output right), giving a straight-through power flow, maximum separation between the noisy input loop and the output terminal, and no chance of a user swapping the two identical connectors... which leads to Q30. 11. **Any indicator LED?** Default: yes, one low-current green LED on VOUT (~1 mA, ~5 mW), as an at-a-glance "output is live" indicator. Cost to efficiency at 3 A is negligible (0.03%). Say no if the pure-efficiency reading is preferred. 12. **Any power-good (PG) output brought out?** Default: no external PG connector; if the chosen IC has a PG pin, bring it to a test pad only. 13. **Test point physical form.** Default: 1.5 mm bare round SMD pads for VIN, SW, FB, EN and GND; the VOUT test point paired with a dedicated scope-ground pad at ~5 mm spacing so a scope spring ground reaches both (this satisfies the brief's "low-inductance scope ground point near the output"). No THT loop test points - they add inductance and height and cost a hand-solder step. 14. **Is EN to be user-controllable (jumper/header), or purely an internal UVLO node?** Default: internal only - a resistor divider from VIN sets UVLO at approximately 6.5 V rising / 6.0 V falling, so the converter refuses to start below the 7 V spec floor, and EN is exposed as a test pad only. 15. **Minimum operating ambient?** Default: 0 C operating, with all parts rated -40 C or lower so the limit is the specification and not the silicon. 16. **Enclosure: is the 50 C figure free-air ambient or inside a sealed box?** Default: open-frame, free-air 50 C, mounted on standoffs with unobstructed convection on both faces. If it is ever boxed, the internal air temperature must be re-stated - a sealed enclosure would likely break the thermal budget. 17. **May the board use the full 50 x 40 mm, or should it be as small as possible?** Default: use the full 50 x 40 mm envelope. Copper area is the free variable that closes the thermal case, and the brief caps the size rather than asking for minimum size. 18. **2-layer or 4-layer?** Default: 4-layer. The brief's own layout section demands "an uninterrupted ground plane on the layer directly under the switching components", a single low-impedance ground reference, and enough copper to sink ~2 W by convection - all three are materially easier and lower-risk on 4 layers, and the JLCPCB cost delta at prototype quantity is small. Justification to be recorded at P5. 19. **Mounting hole geometry.** Default: 3.2 mm holes, non-plated (NPTH), one at each corner inset 3.5 mm from both edges, 6.5 mm diameter copper/mask keepout for an M3 washer, isolated from GND (no chassis-ground bond, avoiding an uncontrolled ground loop through the standoffs). 20. **Maximum component height?** Default: 15 mm above the top surface, which comfortably fits a 5.08 mm screw terminal (~10 mm) and any shielded power inductor likely to be chosen (<7 mm). 21. **Copper weight?** Default: JLCPCB standard stackup - 1 oz (35 um) outer, 0.5 oz inner on 4-layer (the brief explicitly warns to check this). Escalate to 2 oz outer only if the IPC-2152 sizing or the junction-temperature calculation fails at 1 oz; record the escalation if it happens. 22.

Build quantity? Default: 5 boards (JLCPCB prototype minimum). 23. **Target unit cost?** Default: no hard cap; keep the BOM under roughly $12/board at qty 5 and flag any single line item over ~$3 for a conscious decision. 24. **LCSC stock threshold to call a part "available"?** Default: >= 500 pcs in stock for the buck IC, inductor and P-FET; >= 2000 pcs for passives; and no part whose LCSC page shows an end-of-life or "last stock" state. 25. **Assembly method, including the two THT screw terminals?** Default: JLCPCB PCBA for all SMT parts; the two screw terminals hand-soldered on receipt (avoids JLC's THT surcharge on a 5-board run and keeps the connector choice free of JLC's THT part list). Both connector footprints must still be documented in the BOM/CPL as DNP-for-assembly. 26. **Single- or double-sided assembly?** Default: single-sided, top only. All active and passive parts on top; bottom reserved for ground plane, thermal copper and the test pads' ground stitching. This halves assembly cost and keeps the bottom face free to conduct heat. 27. **JLCPCB process class?** Default: the standard/economic class (typically 5 mil (0.127 mm) minimum trace and space, 0.3 mm minimum drill, 0.13 mm annular ring). Nothing in this design needs finer, and staying standard keeps cost and yield where they should be. Actual limits to be confirmed against JLC's published capability table at P5, not from memory. 28. **LCSC Basic vs Extended parts?** Default: prefer Basic/Preferred for R, C, L and the LED; accept Extended for the buck IC, the power inductor and the P-channel FET where the requirement demands a specific part. Extended parts add a per-line feeder fee - acceptable at these quantities. 29. **Required behaviour under a sustained output short circuit?** Default: the board must survive indefinitely and recover automatically when the short is removed (hiccup-mode current limit), without exceeding component ratings and without the input fuse blowing unless a genuine hard failure has occurred. 30. **Required behaviour if the OUTPUT terminal is back-driven, or if input and output connectors are swapped by the user (they are identical 2-pin 5.08 mm parts)?** Default: no active output reverse-current or back-drive protection is required, but the silkscreen must unambiguously mark IN/OUT and polarity on both connectors, and the board should not be destroyed by a brief 5 V back-feed into the output. A swapped connection driving 12 V into the 5 V output is NOT required to be survivable - it will be prevented by labelling. Flag if survivability is actually wanted. 31. **Any EMI standard to meet (CISPR 32 / EN 55032 class B conducted or radiated)?** Default: no formal standard and no compliance testing; apply the brief's layout rules as best practice, leave the DNP snubber footprint as the mitigation hook, and add no input EMI filter beyond the bulk plus ceramic input capacitance already required. 32. **Switching frequency preference?** Default: no hard requirement - choose whatever the selected IC family supports in the roughly 400-800 kHz range, trading inductor size against switching loss and the >88% efficiency floor. If a specific band must be avoided (e.g. AM broadcast, 530-1710 kHz), say so, since that would rule out several otherwise-suitable 1 MHz-class parts. Note the requirement interacts with Q31. 33. **Silkscreen content?** Default: board name "sbuck-5v3a", revision "A", date, input and output voltage/current markings at each terminal, polarity (+/-) on both connectors, pin-1/polarity marks on every polarised part, and every test point labelled. No logo.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Unattended run: orchestrator answered all 33 requirements open questions as delegate (requirements-answers.md)	Brief DECISION POLICY forbids clarifying questions; no mains/battery/motor/>30V/RF flags apply, so no safety unknown is being guessed

Phase	What	Why
P0	4-layer stackup, 1oz outer / 0.5oz inner	Brief demands an uncut GND plane directly under the switching components plus 2.05 W natural-convection cooling on <=50x40mm; 2L cannot give both the image plane and the bottom thermal copper
P0	Add a 4A slow-blow input fuse (DCR <=30 mOhm) beyond the brief's protection list	IC OCP+hiccup cannot protect against a shorted high-side switch; upstream supply current limit is unstated, so the fuse is what bounds fault energy on-board
P0	Target switching frequency 400-700 kHz	Below the AM broadcast band, keeps 18V-input min-on-time and switching loss comfortable, and allows 20-40% ripple with a stocked inductor
P0	Spawn tier deviation: Fable 5 unavailable (out of credits); judgment roles run on Opus instead	Playbook forbids silent downgrade - Opus is the nearest available tier for architect/placement/reviewer roles; recorded rather than hidden
P1	Input fuse: take the 5A slow-blow (Bel 0685T5000-01, 5032 pcs) over the spec-ideal 4A sibling (174 pcs)	Requirements Q24 binds stock >=2000 pcs and the brief says an out-of-stock part is a failed design; 5A still opens on a bolted fault and costs only 119 mW at the 2.44 A worst case
P1	P3 must pull screw-terminal body height, pin diameter and recommended hole size from the vendor datasheet - LCSC attributes do not carry them	Footprint cannot be built or DFM-checked on missing mechanical data; flagged by the protection scout as its open item
P1	All MLCCs must be X7R (no X5R anywhere); Cout bank moves to 22uF/25V 1210 X7R	Architect computed 83-87 C board surface; X5R is rated 85 C so it is a latent failure, and the X7R part also derates less on DC bias (~12% vs ~25%)
P1	Add a 15V Zener + gate resistor clamp on the reverse-polarity P-FET gate	AO4407A survives 18V static on its +/-25V Vgs, but the architect's 25.4V hot-plug ring on the screw terminals exceeds it; the clamp is the cheap fix
P1	In1 and In2 both solid GND; no power net may use an inner layer	0.5oz inner copper needs 3.0-4.6mm widths for these currents per IPC-2152, which is unroutable - so power stays on 1oz outer and the inners are pure return/thermal
P1	Feedback divider uses 0.1% resistors regardless of IC choice	Reference tolerance alone spans -1.2%/+1.8% across the candidates and already eats 60-95% of the +/-2% budget; FB bias current error is negligible (<=0.1%) so the divider tolerance is the only lever left
P1	RC snubber values come from general practice, not a vendor citation - footprint sized generously and left DNP	None of the three candidate vendors publishes a snubber value; the brief wants the footprint anyway, so honest sourcing is to mark it uncited rather than invent a citation
P1	P2 may revisit the 4.0A current-limit-minimum floor - it was a delegate tightening (Q5), not a user requirement	It is the single rule disqualifying LMR33630 (3.85A min, ~3.375A guaranteed Iout = 12.5% over the 3.0A load), and the AP64350 that satisfies it carries a 7-10mA UVLO penalty and external compensation
P2	OVERRIDE of delegate answer Q14: UVLO retargeted from 6.5V/6.0V to 6.2V/5.3V	0.5V hysteresis is less than this board's own 0.49V cable drop at 2.44A (motorboating), and VON=6.5V at the max threshold corner lands ~7.0V so the converter could refuse to start at its minimum rated input; also drops the UVLO divider draw from 81-181mW to 1-2mW

Phase	What	Why
P2	Buck IC: AP64350SP-13 at 500 kHz (RT=200k)	Decided on MAX Rds(on) 75/45 mOhm vs LMR33630's TYPICAL 95/66; TI's own max column gives 1.65W conduction -> Tj ˜112C, over the 105C limit. SY8205 rejected on preliminary datasheet with typ-only Rds
P2	1 oz outer copper; both vendor reference layouts (2 oz) lose	JLC's only 4L/1.6mm 2oz-outer lamination has a 0.4284mm L1-L2 prepreg vs 1080B's 0.2444mm - nearly doubles thermal-via resistance and pushes the image plane 75% further from the hot loop; net +0.8C, so 2oz is worse here
P2	Stackup JLC04161H-1080B; F.Cu + In1 + In2 + B.Cu ALL GND	JLC inner copper is 0.0152mm - thinner than the nominal 0.5oz the rule assumed - so inner power widths would need 3.5-5.3mm; declared explicitly because the pipeline default would otherwise pour In2 as +5V
P2	Single flat root sheet (refdes 1-99, #PWR 100), 33 placed parts	Small board; hierarchy rejected to eliminate the /<sheet>/<LABEL> net-name mismatch class the recipe warns about
P3	R6 feedback top resistor 116k -> 115k (0.1%)	116k is not an E96 value and has zero 0603 stock at any tolerance; 115k/22.1k is what the AP64350 datasheet's own 5V reference design specifies, and 118k was rejected for failing the high corner at 5.130V
P3	U1 ships with NO true pin-compatible alternate (explicit empty list)	Pin-by-pin comparison shows AP64350 and LMR33630 differ on 4 of 8 pins and use different compensation architectures; recording the single-source risk honestly beats listing a fake second source
P3	AO4407A Rds(on) budget checked at the real operating gate drive (Vgs = -Vin, clamped -15V), not at the datasheet's -4.5/-5V row	Gate sits at GND and source at VIN in this topology so Vgs spans -7V to -15V; at the 12V spec point Vgs=-12V gives 13 mOhm max = 26 mW, inside the 22 mOhm budget line. The -5V row (38 mOhm max) is not an operating condition on this board
P3	AO4407A datasheet has no land pattern - librarian uses standard JEDEC MS-012 SOIC-8, 1.27mm pitch	The 4-page Rev3 (2008) PDF carries no mechanical/recommended-land drawing; sourcing the footprint from the JEDEC standard is honest, inventing pad dimensions from the PDF would not be
P3	FB bias-current error treated as an ASSUMPTION, not a datasheet fact - P1 reference-design claimed <=0.1% but the datasheet states no value	Divider current is 0.8V/22.1k = 36 uA, so a typical 10-100 nA FB bias is 0.03-0.3% of it - safe, but it must be labelled an engineering assumption since the vendor never publishes the number
P3	Thermal via array 16 x 0.3mm at 1.0mm pitch is OUR derivation - the vendor spec is only qualitative	AP64350 datasheet says merely 'add as many vias as possible' with an uncounted illustrative array; the numeric array comes from the P1 architect's 1.9 K/W calculation, cross-checked against the Wurth FEA table
P3	Vendor thermal numbers are stated on 2oz copper; board is 1oz - deferred to P8 check_thermal on real geometry	theta_JC is package-internal so it barely moves, and the architect's governing model computed board-to-ambient independently; resolving this on measured copper at P8 beats re-litigating it on datasheet conditions now

Phase	What	Why
P3	U1 exposed pad enlarged from the pulled 3.200x2.500mm to the vendor Suggested Pad Layout 3.502x2.613mm (approved hand-edit)	-12.6% EP area on a design whose theta_JC closes with zero margin and whose entire thermal path runs through that pad; the vendor's own suggested layout outranks the easyeda2kicad pull
P3	F1 library assets come from KiCad stock (Device:Fuse + 1206 fuse land), not an EasyEDA pull	C3163312 is the only LCSC listing for this MPN and its EasyEDA record 404s; re-sourcing would trade a datasheet-verified true slow-blow part for an unverified one, so the library gap is filled instead of the part changed
P3	P4 must take C4 polarity from the footprint silk +/- mark, not the symbol	The pulled aluminium-can symbol has generic pin names 1/2 with no polarity semantics, while the footprint carries an explicit +/- pair with pin1=+
P4	NO series gate resistor on Q1 - my spawn instruction was wrong, the schematic agent correctly followed the architecture	blocks.md B1 and sheets.md s5.4 both explicitly forbid a gate RC, and the P1 architect already proved a gate RC cannot limit inrush here (the body diode is forward in the normal direction); R1=100k is the gate path and D2 clamps Vgs
P4	C3 refitted as the Eq.19 COMP HF pole (COMP->GND) at 10pF, NOT the Eq.20 feedforward across R6; sheets.md s3 to be amended	Measured both: as feedforward it pushes |T| at fsw/2 to -8..-2 dB and breaks the vendor's own <-10 dB gain-margin target; as the Eq.19 pole it costs 8 deg of PM (we hold 65-72 deg) and buys -16 dB
P4	Feedback divider 115k/22.1k -> 105k/20.0k	105k/20.0k hits the required 5.2500 ratio exactly (Vout 5.0000 V) and moves the worst-case low corner from 4.9034 V to 4.9401 V - 3.4 mV of margin becomes 40 mV, at no cost
P4	P9 BOM/CPL must filter DNP by the Variant field manually - no script in the skill reads any DNP mark	J1, J2, R9, C16 are marked Variant=DNP (the only marking a generator can reach, ksa hard-codes (dnp no)); left unfiltered, P9 would ship four parts as populated including both hand-solder screw terminals
P4	DNP snubber upsized: R9 0603->1206, C16 1nF X7R->470pF C0G 1206	Reviewer showed C*V^2*f = 162 mW at 18V (338 mW at the 26V ring) into an 0603 rated 0.1 W - a DNP footprint that destroys itself when populated is a trap; 470pF/1206 gives 76 mW into 0.25 W
P4	Loop calibration claim WITHDRAWN - the x0.83 realisation factor is not evidenced	Reviewer read the vendor's Fig.29 and found C4=33pF IS fitted in the published Bode plot, so the author's 'PM 79-84 vs published 81.6' compared the wrong variant; design still passes at ~46-53 deg worst case vs the 45 deg floor, but the margin is smaller than P4 claimed and only a real loop measurement can close it
P4	Silkscreen connector labelling carried FORWARD to P6/P8 as a tracked requirement	It is delegate Q30's ONLY mitigation for the swapped input/output failure mode (both terminals are identical 5.08mm 2-pin), and it is unreachable from a schematic generator - so it must not die in a P4 note

Phase	What	Why
P4	Hot-plug inrush: 26V voltage declaration KEPT, current bounded by the damped path, residual risk accepted and documented	The 0.05-ohm/180A row assumes an ALL-CERAMIC Cin; this board deliberately carries C4 at 80 mOhm ESR as a damping requirement, so that row is not our configuration. Keeping 26V is conservative for part ratings and costs nothing. Residual: AO4407A body-diode single-pulse capability is unpublished, so the peak cannot be closed from sourced data - and screw terminals are a wire-then-power connector, not a hot-plug one
P4	The clean ERC is WEAK evidence for U1 and must not be cited as pin-level proof	Reviewer noted nearly every U1 pin is declared 'passive' in the pulled library, so ERC cannot catch a mis-wired IC pin here; the real pin-level evidence is the datasheet-extract cross-check and the reviewer's independent netlist read
P4	Load-step prediction restated as 181 mV (not 148 mV) against the 200 mV limit	Reviewer applied the author's own 0.83 realisation factor consistently; also the 0->3A step starts in PFM which the CCM model does not cover, so P8 must treat 9.5% margin as the honest figure
P4	Load step accepted AT the 200 mV limit on the corrected reading (148 mV modelled); R5=75k stands, no rework	Measured R5 sweep shows no value clears fc, PM and dV together - 82k and 91k FAIL phase margin, the binding requirement; and dV is mathematically invariant in COUT (fc scales 1/COUT so fc*COUT is constant), so buying more output capacitance would be pure cost for zero transient gain
P4	The fc 25-50 kHz target was an ORCHESTRATOR proxy, not a requirement - the 24.4 kHz floor is not a miss	The real spec is 100 us recovery; at 24.4 kHz settling is 32.6 us, a 3.1x margin. Recorded so no later phase re-litigates a 2.4% shortfall against a number that was never the specification
P4	Loop compensation is the board's #1 bring-up bench item; R5/C2/C3 are three adjacent 0603 parts, re-tunable without a respin	Every loop number rests on extrapolating a one-point vendor calibration onto a different circuit, and the vendor's published GM matches no reader-built model - only a bench Bode plot can close it
P5	Mounting holes come from the schematic (H1-H4), not board_init - re-run with –mounting-holes 0	Passing 4 duplicated the schematic's own H1-H4 and produced the last 2 parity items; sourcing them from the schematic keeps them real parts with nets and BOM semantics, and P6 places them at the corners
P6	Thermal via array is 12, not the specified 16 - 16 is geometrically impossible	EP is 3.502x2.613mm; four 0.55mm lands at 1.0mm pitch need 3.55mm, and JLC's 0.5mm hole-to-hole floor caps it at 3x4=12 at 0.9-0.95mm pitch. Cost is R_via 2.48 vs 1.90 K/W = +0.5 C of Tj, inside the margin. check_thermal would not have caught this
P6	J1/J2 rotations corrected in placement.edges: J1=270, J2=90 (were 0 and 180)	The declared rotations pointed BOTH screw-terminal wire entries into the board; proven by orthographic side render, because the mouth is the SHORTER plastic face on this part and the WRL bbox alone would have misled
P6	Anneal candidates REJECTED on merit; hand structure kept	Candidates reached HPWL 224.6 vs 234.0 but the 3.1% is almost entirely plane-fed GND, all four carried 5-7 courtyard overlaps after repair, and cand1 pulled R7 3.5mm closer to L1 and lengthened /FB

Phase	What	Why
P6	TOOLCHAIN GOTCHA: placement.fixed silently disables separation constraints	Fixed refs are excluded from the annealer's cluster list, so all 8 separation pairs landed in separation_unknown_refs at zero cost weight; verified all 8 by hand instead (C4-U1 22.2mm, F1-U1 11.1, R6-L1 9.8, J2-L1 16.5) - all pass
P6	P7 must pour the four GND planes AND pour /SW as a zone BEFORE any autoroute	The 0.83 route probe is an artifact: GND is 25 of 71 connections and plane-fed, so unpoured Freerouting tried it as 2.055mm tracks; and aiee_pwr_width_SW (2.31mm) applies to every /SW track including the 0603 bootstrap cap's - a zone is not a track, so the rule does not bind it
P6	P5 omitted the mandatory keepout translation; corrected at P6	placement.keepouts rects were still board-local, so three landed off-board entirely and the fourth produced a phantom violation; translated in constraints.json and annotated
P6	25 structural footprints left LOCKED - P8 fixers must unlock before nudging	Placement cost 8 edit iterations and the hot-loop geometry is the design; locking prevents a later fixer from silently undoing it, but the lock must be visible to whoever needs to move something
P6	thermal.min_vias lowered 16 -> 12 in constraints.json with the geometry recorded	16 cannot physically fit (EP 3.502x2.613mm vs 3.55mm needed at 1.0mm pitch, JLC 0.5mm hole-to-hole floor); 16 was our own derivation not a vendor spec, and leaving it would fail check_thermal at P8 on an unachievable target. Copper-area criteria unchanged, Tj cost +0.5 C
P7	C:/Program Files/Git/SW poured as a ZONE, not routed as tracks	aiee_pwr_width_SW demands 2.31mm on every /SW track including the 0603 bootstrap cap's, which is impossible at that pad; a zone is not a track so the rule does not bind it - the alternative would have been weakening a current-carrying rule
P7	Forced –power-layers In1.Cu,In2.Cu,B.Cu instead of the auto heuristic	auto considers inner layers only, so B.Cu would have been fair game for power routing and would have cut the second radiating face on a board that closes on 7 C of thermal margin
P7	Freerouting necking fixed by pour fan-in; no rule weakened	FR routed at exactly the pad widths (0.8058mm = 0402 pad) despite –pad-window clearing all 61 power pads; remediation ladder step 4 applied, leaving .kicad_dru, .kicad_pro netclasses and constraints.json byte-unchanged
P7	ORCHESTRATOR ERROR corrected by the router: B.Cu >=650 mm2 within 14.3mm of U1 is impossible	That disc clipped to the 50x40 outline is only 608.3 mm2; the constraint's _note says 650 is check_thermal's A_sat SUMMED over layers, which measures 2169.7 mm2. I restated the constraint wrongly in the spawn prompt
P7	route_critical and route_cleanup both skipped, deliberately	route_critical skips plane/pour-carried power nets by design and every power net ended up plane- or pour-carried; route_cleanup's loop-breaker has regressed on pour boards twice and DRC was already 0/0

Phase	What	Why
P8	ACCEPT deletion of 3 of 9 U1 ring vias; U1 keeps 12 EP + 6 ring = 18	Each sat in a 1.06-1.58mm GND finger, so it was current-limited by the finger not the via and carried no real share; the nearest legal >=2.055mm spot was 5-6.5mm away ACROSS the /SW pour, which would route return current under the switch node - exactly what the brief forbids. Deleting beats relocating, and 18 still exceeds the min_vias 12 requirement
P8	SCRIPT DEFECT: check_current.pour_neck erodes each zone fill in isolation	A pour split across zone objects (KiCad knocks a lower-priority same-net zone out of a higher-priority one) reports a phantom 0.00mm neck with pos in a different structure from the real defect; it should union same-net fills per layer or emit a distinct fill_fragmented kind. Cost most of this fix's analysis time
P8	SCRIPT GAP: stitch_vias accepts a candidate whose surrounding same-net pour is narrower than the net's IPC-2152 width	The (60.72,60.225) stitch landed in a 1.06mm GND channel and became a P8 error; existing rejects (foreign_copper, near_same_net_via) do not cover pour width
P8	SELF-CORRECTION: reverted pdn=false on +5V and GND; used waivers instead	pdn=false silenced check_pdn's category error but ALSO excluded both nets from check_irdrop - the two highest-current nets on the board. A waiver states 'this check does not apply' without disabling unrelated analysis. Restored coverage now reads +5V 9.0mV / GND 1.1mV drop
P8	SPICE sim gate NOT run - deliberate scope call, recorded as a deviation	P2 nominated only SIM-1 (FB divider setpoint) and SIM-2 (EN/UVLO thresholds), both static DC-tolerance fragments already verified TWICE analytically with exact numbers by the P4 author and independently by the fresh-context reviewer (who showed 1% parts miss by 32-36mV); house policy forbids simming the buck switching for lack of vendor models, so SPICE would restate arithmetic rather than test anything new
P8	CRITICAL FIX: D2 cathode marker was on the anode end; moved (41.720,59.125) -> (45.900,59.000)	Assembled per the wrong silk, the 15V Zener reverses into a forward diode holding Vgs at -0.7V, Q1 never enhances, 2.6A runs in its body diode at ˜2.1W in an SO-8 - Q1 and the reverse-polarity protection both destroyed. Netlist was always correct; only the marker was wrong. Now 0.640mm from pad 1 (cathode) vs 3.036mm from pad 2, agreeing with the footprint band and pin-1 dot
P8	ACCEPT 3 residual misattributed refdes labels (C9->C7, R7->C12, C2->C3); 8 fixed	At 1.0mm refdes size the inked label is 2.6-3.2mm wide against a 2.5-3.0mm passive pitch, so no attribution-clean position exists at any legal clearance; 0.8mm text would clear two but move_text has no size field. Mitigation: all FUNCTIONAL legends are correct (VIN/VOUT/polarity/TP names), SMT is machine-placed from the CPL not from silk, and only J1/J2 are hand-soldered - their legends are correct

Phase	What	Why
P8	KEEP the fixer's remove_text addition to place_edit.py/place_swig.py (out-of-protocol skill change)	add_text matches on (layer,string,position) so it can create or update in place but never RELOCATE; without remove_text the only options were raw-editing the board (forbidden) or shipping the board-killing D2 marker. Additive, regression-tested, 66/66 place tests pass
P8	THERMAL RESTATED: Tj ~98.7 C, margin 6.3 C (not 7.1), band 87.5-108.4 C at +/-30%	The reviewer showed check_thermal's 0.83 C headroom is NOT a layout property - a_eff measures 2137mm2 and clamps to A_SAT 645, so theta is constant for any 4L board above 645mm2 and the vias/planes move it 0.00 C. The believable number is the P1/P2 ladder, corrected: it still assumed 16 vias at 2.16 K/W, and 12 vias gives ~2.88 K/W = +0.76 C
P9	ORCHESTRATOR DIAGNOSIS WRONG: the TP silk dfm error was NOT a mask-aperture overlap	I estimated a ~0.92mm aperture; measured it is 1.7664mm2 = exactly pi*0.75^2, zero mask expansion. Real cause is lib/gerblib.py read_gerber chording drawn arcs, so KiCad's two 180-degree arcs collapse onto the diameter - a phantom 1.5x0.12mm bar = 0.1798mm2 exactly. My suggested fallback (enlarge the ring) could never have worked, since the chord crosses the centre at any radius
P9	SCRIPT DEFECT (high value, NOT fixed this run): lib/gerblib.py chords drawn arcs	The S12 approximate_arcs fix was applied to _flash_polys only, not to Line/Arc in read_gerber, so EVERY footprint drawing a silk ring around a pad - every stock TestPoint, many polarity rings - will fail the dfm gate forever. Recorded for a future T-step rather than fixed here, because deletion already cleared this board and skill work belongs in its own session
P9	12.7um clearance was a priority-0 pour crumb in a 0.02mm inter-zone lane	KiCad deburrs a pour at min_thickness BEFORE subtracting higher-priority zone outlines, so a sub-min_thickness lane between two zone outlines gets refilled and never deburred; the P8 Z2 edge at y=51.12 'clearing' zone 2 at 51.1 is what manufactured it. Fixed with a priority-6 bridge zone; a second unflagged 0.02x1.93mm hair went with it
P9	DISCLOSE, do not fix: 2 residual etch slivers and a 0.5um island-gap margin	Slivers 0.05x1.71mm and 0.078x0.497mm are fused to neighbouring copper so no gate fires, but a fab may query them; the min F.Cu island gap is 0.1021mm against JLC's 0.1016 floor, so ANY further pour edit risks tripping dfm_clearance - the risk of touching exceeds the risk of leaving

Sheet plan

sbuck-5v3a - schematic sheet plan, net contract and refdes map

1. Sheet plan: ONE flat root sheet

```
| Sheet | Blocks (‘blocks.md‘) | Refdes block | ‘#PWR‘ base |
|---|---|---|---|
| **root** (no child sheets) | B1-B9, all of them | **1-99** | **100** |
```

Decided: a single root sheet, no hierarchy. 33 placed parts (2 of them DNP) plus 7 test pads

and 4 mounting holes. This is one function - a single-rail converter - with one floorplan that runs straight through from the left edge to the right edge, so there is no organising principle for a split to express. The full-run recipe explicitly allows one schematic agent for a board this size.

The decisive argument is the **net-naming risk**, not the part count. KiCad names a net from a hierarchical label as `/<sheet>/<LABEL>`, and every `constraints.json` consumer at P5-P8 matches on the exact string; a mismatch is silently no-op'd rather than failing. That trap cost the lumina-par run a P4 amendment and forced rf-de-20m into a binding root-label contract for a single crossing net. A flat sheet removes the entire class: local labels come out as `/NAME` and power symbols come out bare, with no sheet prefix anywhere.

Refdes are unique by construction on one sheet. The blocks below are a *readability* convention for the BOM, not a uniqueness mechanism.

—

2. Net-naming contract - BINDING ON P4

These are the CANONICAL names. `constraints.json` matches them literally.

```
| Canonical net | Kind | Carries |
|---|---|---|
| `/VIN` | local label | J1.1 -> F1 -> Q1 drain.  **2.44 A at 7 V**, raw and
unprotected |
| `+VIN` | **power symbol / global label (bare)** | Q1 source -> C4-C9 -> U1
VIN. 2.6 A design, 1.5 A rms in the ceramics |
| `/SW` | local label | U1 SW -> L1.  **3.53 A pk, hard-switched, 500 kHz** |
| `+5V` | **power symbol / global label (bare)** | L1 -> C10-C15 -> J2.  3.3 A
design ceiling |
| `GND` | **power symbol (bare)** | F.Cu island + In1 + In2 + B.Cu.  3.3 A
return |
| `/EN` | local label | R2/R3 junction -> U1 EN. TP5 |
| `/FB` | local label | R6/R7 junction -> U1 FB. High-Z, keep away from `/SW`
and L1 |
| `/COMP` | local label | U1 COMP -> R5 -> C2 -> GND |
| `/BST` | local label | C1 -> U1 BST |
| `/RT` | local label | U1 RT -> R4 -> GND |
| `/QGATE` | local label | Q1 gate, R1 to GND, D2 anode |
```

Rules P4 must not break:

1. `+VIN`, `+5V` and `GND` must be **power symbols or global labels** so they come out bare. A local label would produce `/+VIN` and five `constraints.json` entries would stop matching in silence. 2. Every other net above must be a **local label on the root sheet**, producing the leading `/`. Do not "tidy" the slash away. 3. **Do not rename anything to end in `_P`, `_N`, `_H`, `_L`, `DP`, `DM`, `+` or `-`.** `rules_gen.detect_diff_pairs` auto-pairs `high_speed` nets with those suffixes; `/SW` is declared under `high_speed` and a suffix collision would silently emit differential-pair gap rules and an inner-layer track ban on the most current-carrying node on the board. Current names are safe. 4. `diff_pairs` is an **explicit empty list** in `constraints.json` - there is no differential net on this board and the empty list disables auto-discovery.

—

3. Refdes map

B1 input entry + reverse polarity (`/VIN`, `/QGATE`)

```
| Ref | Part class | Note |
|---|---|---|
| **J1** | 2-pin 5.08 mm THT screw terminal, DB128L-5.08-2P class | **LEFT short
edge.** DNP for assembly, hand-soldered (Q25) |
```

F1	5 A slow-blow 1206 chip fuse, Bel Fuse C1T class	63 V DC, 20 mOhm, 5.3 A^2s. >= 20 mm^2 copper per pad, kept off the U1/L1 zone
Q1	P-channel MOSFET SOIC-8, AO4407A class	**drain to input, source to load.** >= 50 mm^2 of drain+source copper
R1	100k 0603	gate to GND. Also the polarity-reversal turn-off path
D2	15 V Zener SOD-123	anode at gate, cathode at source. Clamps |Vgs|

B2 input capacitance (+VIN)

Ref	Part class	Note
C4	100 uF / 35 V hybrid-polymer alu SMD can, KNM2100UF35V149EC0055 class	**ESR 50-300 mOhm is a REQUIREMENT.** Place at the input edge (coolest corner). Not a low-ESR polymer, not a plain electrolytic
C5-C8	4x 4.7 uF / 50 V X7R 1206, 1206B475K500NT class	1.5 A rms shared four ways
C9	100 nF / 50 V X7R 0603	**AT the U1 VIN pin.** Innermost element of the hot loop

B3/B4 converter, compensation, UVLO (+VIN, /SW, /EN, /FB, /COMP, /BST, /RT)

Ref	Part class	Note
U1	AP64350SP-13 class, SO-8-EP	Single GND pin + exposed pad on the same net. No VCC pin
C1	100 nF / 50 V X7R 0603	BST to SW. Required, not optional
R4	200k 1% 0603	RT -> 500 kHz (‘RT[kOhm] = 100000/fsw[kHz]‘)
R5	Rcomp, 0603	vendor start 14k. **P4 RE-DERIVES against 5x22 uF**
C2	Ccomp, 0603	vendor start 3.3 nF. **P4 re-derives**
C3	10 pF C0G 0603	**AMENDED at P4.** Eq.19 COMP HF pole, COMP to GND - NOT the Eq.20 feedforward across R6. Measured both: as a feedforward it pushes |T| at fsw/2 to -8..-2 dB and breaks the vendor’s own <-10 dB gain-margin target; as the Eq.19 pole it costs 8 deg of phase margin (the loop holds 65-72 deg) and buys -16 dB. No net or placement-group impact
R2	**105k 1% 0603**	+VIN -> EN. UVLO top
R3	**24.0k 1% 0603**	EN -> GND. UVLO bottom. Target VON 6.2 V / VOFF 5.3 V

B5/B6 output filter and feedback (/SW, +5V, /FB)

Ref	Part class	Note
L1	6.8 uH molded alloy-composite, FAUL1050-6R8MT class	11.5 x 10 mm, 4.1 mm. **>= 125 C rated, DCR <= 25 mOhm at 20 C**
C10-C14	5x 22 uF / 25 V X7R 1210, TCC1210X7R226K250MT class	**4x is the authorised floor, 3x is not** (blocks.md B5)
C15	4.7 uF / 16 V X7R 0805	output HF bypass at the terminal / test point
R6	~116k **0.1%** 0603	FB top. 0.5% is the absolute floor
R7	~22.1k **0.1%** 0603	FB bottom. Ratio 6.249 -> 5.000 V from a 0.8 V ref

B7/B8/B9 indicator, output, snubber

Ref	Part class	Note
D1	green 0805 LED, KT-0805G class	~1 mA
R8	2.2k 0603	LED series

```
| **J2** | same class as J1 | **RIGHT short edge.** DNP for assembly |
| **R9** | 0603, **DNP** | snubber R. 10-33 Ohm if ever fitted |
| **C16** | 0603, **DNP** | snubber C. 470 pF - 2.2 nF if ever fitted |
| **TP1-TP7** | 1.5 mm bare SMD round pads | '+VIN', '/SW', '+5V', '/FB', '/EN',
'GND', and a scope-ground pad ~5 mm from TP3 |
| **H1-H4** | 3.2 mm NPTH | corners, inset 3.5 mm, 6.5 mm keepout, isolated from
GND |
```

Total: 33 placed components (2 DNP), 7 test pads, 4 holes.

—

4. Floorplan for P6 - straight-through, left to right

The board is 50 mm wide (x) by 40 mm tall (y); the **short edges are left and right**, which is where the two connectors go (Q10).

```
| Zone | x | Contents |
|---|---|---|
| **input** | 0 - 13 | J1, F1, Q1, D2, R1, C4.  The coolest corner - C4's life
is set by local board temperature |
| **converter** | 13 - 32 | C5-C9, U1 (thermal island, centred near x=21 y=20),
C1, R2-R5, C2, C3, /SW pour, R9/C16, L1 |
| **output** | 32 - 50 | C10-C15, R6/R7, D1/R8, TP1-TP7, J2 |
```

Non-negotiables that no annealer can discover:

1. **Hot loop first.** `C5-C9 -> U1 VIN -> U1 PGND -> C5-C9` is the shortest loop on the board, all on F.Cu, with C9 physically closest to the pin and In1 solid GND directly beneath as the image plane. 2. **'/SW' is a short wide pour**: >= 2.5 mm wide, <= 8 mm long, <= 40 mm^2, F.Cu only, never inner, never near a board edge or connector. The TP2 tap is a stub that counts against the 40 mm^2. 3. **R6/R7 and the '/FB' trace hug the FB pin and stay away from '/SW' and L1** - an explicit rule in all three candidate vendor layouts. 4. **R9/C16 sit directly across the SW pour to PGND** with a low-inductance path even while unpopulated. Place by explicit `place_edit`. 5. **C4 and F1 stay >= 10-12 mm from U1 and L1** (declared as `separation` entries): C4 because its life halves every 10 C, F1 because it is a deliberate 0.15 W heat source. 6. **16 thermal vias in U1's pad** at 0.3 mm / 1.0 mm pitch, plus 8-12 stitching vias in the surrounding F.Cu island. `check_thermal` will not catch their absence (see `power_tree.md` s4.3) - this is a review gate. 7. **No layer transitions on '/VIN', '+VIN', '/SW' or '+5V'.** A transition needs 6-8 vias at 0.5 A/via. GND is plane-fed through the stitching field.

Parts to place by explicit `place_edit` and **lock** before the annealer runs: `U1`, `L1`, `C5-C9`, `C1`, `R9`, `C16`, `J1`, `J2`, `H1-H4`.

—

5. P4 notes (schematic phase)

1. **Re-derive the compensation.** The vendor's Rcomp/Ccomp are quoted for 2x 22 uF; the bank is 5x 22 uF. Target fc 25-50 kHz, phase margin >= 45 deg, using the AP64350 datasheet's Eqs 12-20. Record the numbers on the schematic. If the loop cannot close there, drop to 4x 22 uF and record it. 2. **R6/R7 must be 0.1% parts in the BOM**, not 1% with a 0.1% note. The reference tolerance alone spends 60% of the +/-2% window. 3. **UVLO is 6.2 V / 5.3 V, not the delegate's 6.5 / 6.0.** Use the AP64350 divider equations (they include the 4.114 uA and 5.5 uA internal pull-ups - a naive divider calculation is wrong) and verify against the P8 sim bench. 4. **No gate RC on Q1 and no gate capacitor.** The body diode already bypasses the channel during inrush. This is a decision, not an omission - do not let ERC or a reviewer add one. 5. **The snubber R9/C16 must exist in the netlist** as DNP so their pads, clearance and routing are accounted for at P6/P7. Populating later is then a BOM change, not a respin. 6. **Test pads are real nets.** TP7 is a second GND pad placed for a scope spring ground, ~5 mm from TP3, not a duplicate of

TP6. 7. **Silkscreen** (Q33): board name + rev A + date; "VIN 7-18V" and "VOUT 5V 3A" with polarity marks at both terminals - this is the *only* mitigation for a swapped-connector event, which is not survivable by design; pin-1 marks on every polarised part; every test point labelled; no logo. 8. **J1 and J2 are DNP for assembly** and must appear as such in the BOM/CPL.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 2 attempt(s); last 2026-08-09T02:14:15, failing 0 of 0.

Review waivers

ERC / schematic-review waivers - sbuck-5v3a P4

ERC gate: **0 errors, 0 warnings** (kc.py erc, exit 0). Schematic review (fresh context): **0 errors, 7 warnings**. No netlist change was required by any finding. Full detail: `reports/review-schematic.md` + `.json`.

Two warnings were FIXED rather than waived (see below). Five are waived here.

—

FIXED, not waived

F1. DNP snubber overstressed its own resistor. C*V^2*fsw = 162 mW at 18 V and 338 mW at the declared 26 V ring, into an 0603 rated 0.1 W. A DNP footprint that destroys itself when populated is a trap, not a feature. Fixed: R9 0603 -> 1206 (>= 0.25 W), C16 1 nF X7R -> 470 pF C0G 1206. New figures 76 mW at 18 V, 159 mW at 26 V. Both values stay inside the documented 10-33 ohm / 470 pF-2.2 nF bring-up range from blocks.md B9.

F2. Loop calibration claim withdrawn. P4 claimed its model was validated against the vendor Bode plot ("PM 79-84 deg vs published 81.6"). The reviewer read Fig.29 and found the plotted circuit HAS C4 = 33 pF fitted, so the comparison used the wrong model variant. The x0.83 realisation factor is therefore unevidenced. The claim is withdrawn from the record; the design is re-stated on its worst reading.

—

WAIVED

W1. Loop phase margin is less certain than P4 claimed. Reviewer's independent re-derivation reproduces P4's fc/PM/|T| table to the digit, so the arithmetic is right; what fails is the calibration against the vendor plot. Worst-case reading gives PM ~46-53 deg against the >= 45 deg requirement. *Waived because*: it still meets requirement on every reading, and the vendor's own published GM (-26.8 dB) matches neither model variant (-8.3 with feedforward, -18.4 without), i.e. the vendor's "first-order calculated" model carries HF poles that no reader-built model reproduces. Closing this needs a real loop measurement or the vendor's own model - not more reading. **Carried to P8 as a bench item.**

W2. Load step: 148 mV modelled, 200 mV on the corrected reading - AT the limit. SUPERSEDED TWICE, and the honest number got worse each time. The first restatement (181 mV) still carried the withdrawn 0.83 factor. Re-running the vendor comparison like-for-like - model WITH C4=33pF and C6=47pF fitted, matching the circuit actually plotted in Fig.29 - gives a factor of 0.743, not 0.83, and the model is optimistic on all three axes (fc +35%, PM +19.4 deg, GM +18.5 dB). Applying those offsets: **dV = 200 mV against a 200 mV limit.** Additionally the 0->3 A step STARTS IN PFM (automatic PWM/PFM, no MODE pin) which the CCM model does not describe.

Waived - as an at-limit DISCLOSURE, not a pass - because: 1. **No R5 value clears all three targets.** Measured sweep, recorded in root.py s2.5a so the "just raise R5" reflex is answered in the source: 75k -> PM 45.6 deg / dV 200 mV; 82k -> PM 44.1 deg FAIL / dV 183 mV; 91k -> PM 39.9 deg FAIL / dV 165 mV. Raising fc buys sampling-pole lag. 75k is the only value that keeps phase margin - the binding requirement - above its floor. 2. **Adding output capacitance does not help.** fc scales as 1/COUT at fixed compensation, so fc$COUT$ *is constant and* $dV = dI/(2\pi fc\mathrm{COUT})$ is INVARIANT in COUT. Buying more caps would be pure cost for zero transient improvement. 3. **It is not a measurement.** It extrapolates a one-point vendor calibration, taken at a different operating point, onto a different circuit. It is a plausibility band, not a second reading. 4. **The dominant uncertainty is larger than the shortfall.** The COUT derating band is +/- 13% around its mean with no vendor DC-bias curve behind it. 5. **Re-tunable without a respin**: R5, C2 and C3 are three adjacent 0603 parts.

Carried to P8 and to bring-up as the board's #1 bench item.

W2b. P8 UPDATE - the phase-margin floor is NOT actually bounded, and this disclosure must stop reading as "0.6 deg clear". The P8 fresh-context review accepted the dV COUT-invariance argument but found the PM argument incomplete: PM is NOT invariant in COUT, and the [75.0, 96.8] uF band is **DC-bias-only**. Hot X7R derating and aging are named in root.py as pushing toward the low end but were never STACKED. Stacked, the effective bank is ~62.7 uF - about 20% below the swept corner - which on the sweep's own slope costs ~3 deg and puts **PM at roughly 42-44 deg, under the 45 deg floor**. No respin is implied (R5/C2/C3 are three 0603 parts and this is already bring-up bench item #1), but the honest statement is that the loop margin at the true COUT floor is UNBOUNDED BY ANALYSIS, not comfortably clear. Closing it needs a network analyser, not more modelling.

W2a. The fc floor of 24.4 kHz is NOT a requirement miss. The "fc 25-50 kHz" target was the ORCHESTRATOR's proxy, written into the P4 spawn prompt as a stand-in for the real requirement, which is "recovery within 100 us". At fc = 24.4 kHz settling is $5/(2\pi fc)$ = **32.6 us, a 3.1x margin on the 100 us spec**. The proxy was simply stricter than the requirement it stood for. Recorded so that no later phase re-litigates a 2.4% miss against a number that was never the spec.

W3. Hot-plug inrush peak cannot be bounded from published data. The 57 A figure comes from the 0.30 ohm path row; the 26 V node rating comes from the 0.05 ohm row of the same table, which assumes an ALL-CERAMIC Cin. At 0.10 ohm the formula gives 180 A against Q1's -60 A channel IDM. *Waived because*: this board deliberately does NOT have an all-ceramic Cin - C4 carries 80 mOhm ESR as a stated DAMPING requirement - so the 0.05 ohm row is not this configuration. The 26 V declaration is kept anyway as conservative headroom for part ratings. Residual and honest: AO4407A's body-diode single-pulse capability is not published, and screw terminals are a wire-then-power connector, not a hot-plug one. **Recorded in DECISIONS.md as an accepted, quantified risk.**

W4. Q1 gate dv/dt immunity is unbounded. parts/C16072.json publishes Qg and Rg but no Ciss/Crss/Coss for the AO4407A. *Waived because*: the datum does not exist in the sourced datasheet, so no analysis can close it. Stated as a limit of knowledge rather than assumed benign.

W5. UVLO effective anti-motorboating margin is 0.32 V, not the 0.41 V argued at P2 (the fuse and FET drops were omitted from that argument). *Waived because*: the margin is still POSITIVE, and the raw hysteresis (0.905 V, VON 6.230 / VOFF 5.325, verified exactly against the vendor's pull-up-inclusive Eq.2/3) comfortably exceeds the 0.49 V cable drop that motivated the P2 override. Worst-case VON 6.67 V sits 0.33 V below the 7.0 V minimum input, so the converter starts across its whole rated range.

—

Non-violations the reviewer flagged for the record

- **The clean ERC is weak evidence for U1.** Nearly every U1 pin is declared `passive` in the pulled library, so ERC structurally cannot catch a mis-wired IC pin here. The real pin-level

evidence is the datasheet-extract cross-check plus the reviewer's independent netlist read. Do not cite "ERC 0/0" as pinout proof.

- The schematic plot is stub-label style, so bring-up needs the netlist alongside it.
- **Silkscreen connector labelling is unreachable from a schematic generator** but is delegate Q30's ONLY mitigation for a swapped input/output connection (both terminals are identical 5.08 mm 2-pin parts). **Carried to P6/P8 as a tracked requirement so it cannot die in a P4 note.**

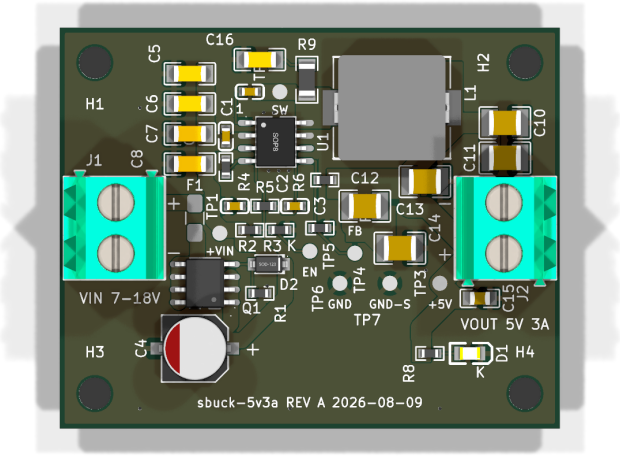
Unresolvable from sourced data (stated, not assumed away)

- No vendor DC-bias curve exists for ANY MLCC on this board, so the [75.0, 96.8] uF COUT band - and therefore every loop number above - rests on conventional estimates. This is why COUT was closed over a band rather than a point.
- No datasheet exists for D1 (KT-0805G), so Vf = 2.8 V is unverified. Bracketed: 0.82-1.36 mA over any plausible green Vf. Outcome is insensitive.
- No vendor snubber value is published by any of the three candidate ICs.
- FB input bias current is unpublished (irrelevant at a 40 uA divider current).

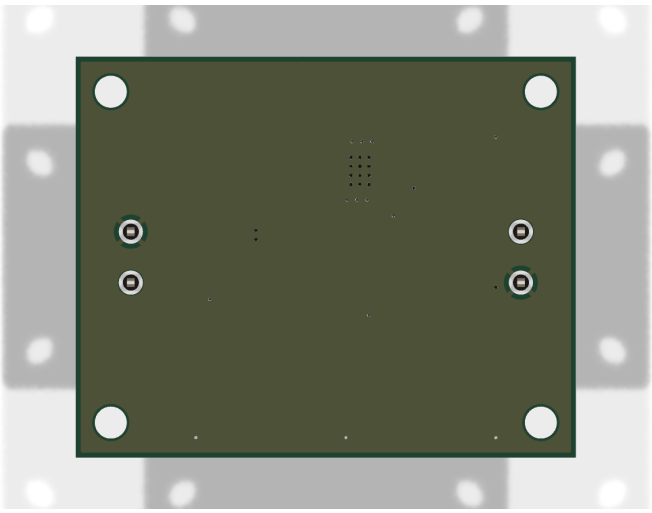
reports/schematic.pdf not available.

6 Layout

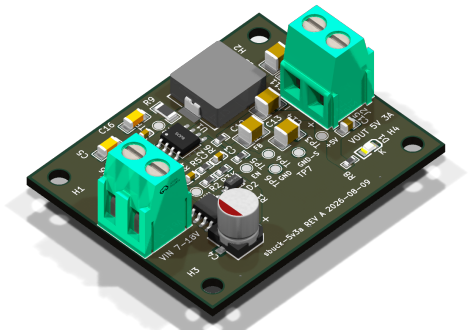
Gate `place` (phase P6): **pass** after 1 attempt(s); last 2026-08-09T03:22:36, failing 0 of 0.
Gate `drc_routed` (phase P8): **pass** after 4 attempt(s); last 2026-08-09T06:30:36, failing 0 of 0.



`reports/renders/sbuck-5v3a_top.png`



reports/renders/sbuck-5v3a_bottom.png



reports/renders/sbuck-5v3a_iso.png

7 Verification

Gate `verify` (phase P9): **pass** after 4 attempt(s); last 2026-08-09T06:54:58, failing 0 of 16.

Verification checks

Check	Status	Findings	Note
check_return_path	skipped	0	skipped - no constraints
check_current	skipped	0	skipped - no constraints
check_decoupling	skipped	0	skipped - no decoupling
check_diffpair	pass	0	
check_creepage	skipped	0	skipped - no constraints
check_thermal	skipped	0	skipped - no constraints
check_silk	violations	3	
check_pdn	skipped	0	skipped - no constraints/decoupling

Design review of record

P8 board review - sbuck-5v3a (fresh context, adversarial)

Reviewer did not place, route or fix this board. Every geometric number below was re-measured from `kicad/sbuck-5v3a.kicad_pcb` with pcbnew + shapely, not read from a prior phase's report. Renders regenerated at 2400 px: `reports/renders/sbuck-5v3a_{top,bottom,iso}.png`.

Gate: 2 errors, 12 warnings. Both existing waiver groups upheld (9 findings), one with a corrected justification. Two new items recommended for documented-deviation status (W1 `/SW` length, W12 mounting-hole keepout).

Independent re-runs (my own, not inherited):

- `verify_all` reproduces the gate exactly: 28 total, 9 error (all waived), 19 warning. Nothing was SKIPPED for missing inputs.
- `kc.py drc --parity --all-track-errors`: **0 violations, 0 unconnected**. Confirmed.
- `check_irdrop` / `check_pdn_z` / `check_thermal` re-read; arithmetic re-derived below.

—

ERRORS

E1. D2's cathode marker `K` is printed at the anode end - `F.SilkS`, `(41.720, 59.125)` The board-level `gr_text "K"` added at P6 sits at x 41.72. D2 is at (43.620, 60.825) rot 180:

	board x	net	role
pad 1	45.26	‘+VIN‘	**cathode**
pad 2	41.98	‘/QGATE‘	anode

Evidence the cathode is pad 1, three ways: the symbol (`lib/aiee.kicad_sym`, BZT52C15) declares pin 1 name `K`, pin 2 name `A`; the footprint (`lib/aiee.pretty/SOD-123_L2.7-W1.6-LS3.7-RD.kicad_mod`) draws its band at local x -0.97 and its pin-1 dot at local (-1.85, 0.80), **both on the pad-1 side**; and the netlist puts pad 1 on `+VIN`, which is where the cathode of a Vgs clamp belongs.

The `K` text is 0.26 mm from pad 2's x-span (it sits directly above the anode pad) and 3.54 mm from the cathode. **It contradicts the same part's own band and dot.** Two conflicting cathode indications is worse than none.

Consequence, if a board is hand-built or reworked to the silk: D2 goes in backwards and becomes a plain forward diode from `+VIN` to `/QGATE`, holding Vgs at about -0.7 V. AO4407A's Vgs(th) is -1 to -3 V, so Q1 never enhances and the whole 2.6 A input current runs in its body diode - roughly 0.8 V x 2.6 A = **2.1 W in an SO-8 with no heatsinking**. Q1 dies and the reverse-polarity protection goes with it.

The circuit is correct; only the marker is wrong. Fix: `silk_place` the text to ˜(45.6, 59.1), or delete it and let the footprint band do the job. **domain: silk.**

Cross-check that this is an isolated slip, not a systematic one: D1's `K` at (64.770, 71.625) IS correct - `LED0805-R-RD` puts pad 1 at local +1.05 (anode, net `/LEDA`) and pad 2 at -1.05 with the chamfered cathode bracket, and the marker is at pad 2. Only D2 is wrong.

E2. Eight refdes labels are attributed to the wrong part - `F.SilkS` These are `check_silk`'s eight `silk_misattributed` **warnings**. They were neither fixed nor waived - they simply fell through the P8 gate, which only fails on `error`. They are real, and three of them are worse than "misattributed":

- **‘R5‘ / ‘C2‘ / ‘R3‘ print as a left-to-right row above the parts C2 / R5 / C3** (x 40.22 / 43.22 / 46.22, all at y 55.025). Every label in that row names a different part from the one beneath it. `C2`'s text at (44.995, 53.025) is 0.47 mm from C3; `R3`'s at (46.72, 53.318) is 0.27 mm from R6 and 1.7 mm above C3, while the real R3 is 4.6 mm away at (44.720, 57.425). Values involved: C2 = 3.3 nF, R5 = 75 k, C3 = 10 pF, R3 = 24.0 k - four different parts, three swappable 0603 bodies. Visible directly on `reports/renders/sbuck-5v3a_top.png`.
- **‘C9‘ at (35.695, 48.060) prints ON C7** (0.00 mm). C9 is the 100 nF 0603 that sets the hot loop; C7 is a 4.7 uF 1206.
- **‘R7‘ at (53.298, 55.725) prints ON C12.**

Plus Q1->C4 (0.65 mm), TP1->F1 (0.29 mm), TP2->R9 (0.25 mm), TP7->TP6 (0.70 mm).

Why this is an error and not cosmetics on this board specifically: parts/parts.json records J1/J2 as **DNP for JLC assembly, hand-soldered on receipt**, so a human is building part of this board from the silk; and TP1/TP2/TP7 mislabel probe points on a live 18 V converter.

The **functional** test-point legends are fine and correctly placed - +VIN, SW, EN, GND, GND-S, FB, +5V each sit adjacent to their own pad (verified on the render). This finding is about the refdes layer only.

Scripted fix, no copper touched: place_edit.py move_text / silk_place.py. **domain: silk.**

—

WAIVER AUDIT - the nine waived findings

Waiver group 1: 7 x check_silk "silk circle covers pad TPn.1 (1.77 mm2)" - UPHELD Verified on three independent legs. The first two are decisive; the third I had to prove rather than accept.

(a) The geometry, read straight out of the board file. All seven TPs carry:

```
(fp_circle (center 0 0) (end 0 0.95) (stroke (width 0.12) (type solid)) (fill
no) (layer "F.SilkS"))
(pad "1" smd circle (at 0 0) (size 1.5 1.5) (layers "F.Cu" "F.Mask"))
```

pcbnew reports GetFillMode() = 1, and pcbnew.FILL_T_NO_FILL == 1 (KiCad's enum starts at 1; FILLED_SHAPE == 2). Ring inner edge = 0.95 - 0.12/2 = **0.89 mm**; pad radius = **0.75 mm**; clearance **0.14 mm**. The claim is exactly right.

(b) The checker bug, located. check_silk.py:167-172:

```
if kind.endswith("circle"):
c, e = _nums(_kid(node, "center")), _nums(_kid(node, "end"))
r = math.hypot(e[0] - c[0], e[1] - c[1])
return Point(xf((c[0], c[1]))).buffer(r + w2) # <- FILLED disc
```

_shape_geom never reads the (fill ...) node at all, for any shape kind. So the ring becomes a filled disc of radius 0.95 + 0.06 = 1.01 mm, which swallows the pad whole. Reported overlap 1.7659 mm2 vs pi*0.75^2 = 1.76715 - a 0.07% shortfall that is just shapely's polygon approximation of a circle. Exactly the signature the waiver claims.

(c) The DRC cross-check is valid here, but its stated generality is not. I did not take "kicad-cli DRC reports 0 violations" on trust. On a scratch copy I shrank TP4's silk ring to r = 0.30 - **entirely inside** the 0.75 mm pad - and 'kicad-cli pcb drc –severity-all' still reported **0 violations**. I then injected two fat silk segments **crossing** TP3, U1.2, U1.9 and C9's pads and got 5 x silk_over_copper + 9 x silk_overlap warnings. So:

> KiCad's silk_over_copper fires on silk that crosses a pad's mask aperture, and stays > silent on silk fully enclosed inside one.

A ring at r = 0.95 overlapping a 0.75 mm pad would be a *crossing*, so DRC's silence on this board **is** real evidence and leg (c) stands. But the waiver's wider phrasing - "the playbook itself directs that silk be verified with kc.py drc rather than check_silk" - is too broad and should be narrowed in the record: kc.py drc is not a general substitute for check_silk, and this board has no working automated check for enclosed-silk cases.

Verdict: waiving the checker, not the geometry. Correct. Keep all seven.

Waiver group 2: 2 x check_pdn "+5V / GND has no decoupling capacitors" - UPHELD, but the stated reason is wrong in a way that matters The **call** is right. The **reason** is not, and the difference changes what the human should be told.

check_pdn does not model "a supply rail must have decoupling capacitors near its IC pins". Reading the source: rail_caps() filters decoupling.json["associations"] by rail == <net>

and raises an error if that list is empty. **It never looks at the board.** Near-pin geometry is `check_decoupling`'s job, and that check passed on its five `+VIN` associations (loop 5.05-11.50 nH).

So the finding is not a category error about IC pins. It is a literal and true statement about the project's own metadata: **no `decoupling.json` association names `+5V`, so no automated check has ever looked at the output bank.** "Nothing is hidden" overstates it.

What I verified about the rails themselves, so the human knows the substance is sound:

- The `+5V` bank exists and is connected: C10-C14 (5 x 22 uF 1210) and C15 (4.7 uF 0805) all carry a `+5V` pad and a `GND` partner; `+5V` F.Cu is **one 198.93 mm2 island**; `kc.py drc --parity` is clean, so schematic and board agree.
- `check_irdrop` DID run on both nets with the real geometry (they were correctly NOT given `pdn:false`): `+5V` worst drop **8.96 mV**, `GND` **1.15 mV**, against a +/-100 mV output window. This is the leg that actually covers these two rails.
- `GND` is a return net; decoupling to itself is meaningless. Correct.

Verdict: keep both waivers, but restate the reason as "check_pdn is a metadata inventory and `+5V` has no association declared; coverage for this rail comes from check_irdrop and P4 arithmetic, not from a PDN check". Also fix `decoupling.json`'s `_note`, which still claims `+5V` is handled by `pdn=false` - `constraints.json` records that as reverted.

—

THE SPECIFIC CLAIMS, RE-MEASURED

claim	stated	my measurement	verdict
vias inside the hot-loop region x 38.87-43.02 / y 46.90-49.20	0	**0**. Nearest are the EP thermal array at x 43.42, outside the region and inside the return path anyway	confirmed
In1 GND under that region	unbroken image plane	**100.00%** coverage - 9.545 of 9.545 mm2	confirmed
‘/SW‘ copper	30.4431 mm2, F.Cu only, <= 40, >= 2.5 mm wide	**30.4422 mm2**, F.Cu only, **1 island**; conduction path U1.8 -> L1.1 survives a **1.5 mm** erosion, so it is **>= 3.0 mm wide throughout**	confirmed (area + width)
‘/SW‘ length	<= 8 mm	**10.490 mm**	**exceeded - see W1**
‘/SW‘ near an edge or connector	no	5.704 mm to the outline, 16.63 mm to J2, 17.22 mm to J1	confirmed
‘/SW‘ inner-layer copper	none	**none** - the board has 28 vias total and every one is GND; every non-GND net is F.Cu-only	confirmed
FB senses at C12’s ‘+5V‘ pad	~2.21 mm, not the inductor pin	R6.1 taps the ‘+5V‘ node at (51.700, 52.997); copper path R6.1 -> node = 1.97 mm, node -> C12.1 = 2.03 mm. L1.2 is **13.3 mm** away. The divider taps the same node C12 does	confirmed
FB routed away from ‘/SW‘ and L1	yes	FB copper -> SW copper **2.935 mm**; FB copper never overlaps L1’s body (FB x 48.27-48.58, L1 body starts x 50.17 -> ~1.6 mm horizontal); R6<->L1 9.835 mm and R7<->L1 13.994 mm against the 5 mm constraint	confirmed
U1 EP thermal vias	12	**exactly 12**, 3 cols x 4 rows, **0.9 mm pitch both axes**, 0.3 mm drill -> **0.600 mm** hole-to-hole against JLC’s 0.5 floor	confirmed
16 geometrically impossible	yes	4 columns inside the EP’s **2.613 mm** need pitch <= (2.613-0.6)/3 = **0.671 mm** -> 0.371 mm hole gap, **under** the 0.5 floor. 4 at 1.0 mm pitch needs 3.55 mm. Impossible either way	confirmed
ring vias	6 (3 of 9 deleted)	**6**, at y 45.700 and y 51.500, x {43.1, 44.1, 45.1} and {43.6, 44.6, 45.6}	confirmed

```
| In1 solid GND | 444.4 mm2 within 12 mm, one piece, no split | **444.92 mm2**
within 12 mm; **ONE island of 1850.07 mm2**; 14 voids, every one of them a
mounting hole or a J1/J2 THT antipad, **none within 12 mm of U1** | confirmed |
| F.Cu GND island at U1 | >= 100 mm2, contiguous with the EP | **1340.60 mm2**,
contains the EP | confirmed, 13x over |
| B.Cu GND | >= 1500 mm2 of 2000 | **1850.07 mm2 = 92.1%** of the 2009 mm2
outline | confirmed |
| L1 '+5V' pour | grown to 153.3 mm2 | **153.23 mm2** within 14.3 mm | confirmed
|
| J1/J2 wire entry | off-board | **confirmed on the iso render**:  J1 entry -x,
J2 entry +x, screws top-accessible, bodies 0.325 mm inside their edges |
confirmed |
```

Was deleting 3 of the 9 ring vias right? Yes, on the evidence. The ring vias sit 1.12-1.17 mm outside the EP edges and serve secondary spreading; the 12-via EP array is the primary junction-to-plane path and is untouched. The stated reason (they sat in 1.06-1.58 mm GND fingers) is the right kind of reason, because a 0.6 mm via in a 1.06 mm finger leaves 0.23 mm each side. And the check that would catch the consequence is armed: `GND` is declared `plane_fed: true`, which keeps **pour-neck findings at ERROR severity at the full 3.3 A budget**, and `check_current` reports **zero** pour-neck errors - only the ten leaf-tap advisories. Thermal cost of 9 -> 6 in the ring is second-order against the EP array.

The mounting-hole keepout that was never built `constraints.json` placement note (3) asked for a 6.5 mm mask/copper keepout at each M3 hole, hand-added as rule areas after the pours. **There are no rule areas on this board.** I checked the consequence rather than assuming it: all four layers are GND at every hole (In1/In2/B.Cu each void a 3.7 mm circle around the 3.2 mm NPTH, so no inner copper is exposed at the barrel), and the nearest **non-GND** copper to any hole is **5.38 mm** (H2, F.Cu) against an M3 washer radius of 3.25 mm - **2.13 mm clear**. Benign. Record it as a knowingly-dropped constraint rather than leaving it silently unmet.

—

U1's THERMAL NUMBER - which model to believe

Do not report "54.17 C against 55 allowed, 0.83 C of headroom" as this board's tightest number. It is not a measurement of this board at all.

`check_thermal` computes 'a_eff = min(A_SAT = 645, sum over layers of heatsink-net copper within 14.3 mm). `I measured that sum: **2137.56 mm2**. So a_eff'` **clamps** to 645, and

```
theta = 45 + 95 * exp(-645/235) = 45 + 95*0.06427 = 51.1055 C/W
rise = 1.06 * 51.1055 = 54.17 C
```

At the clamp, `theta` is a **universal constant** for any 4-layer board with >= 645 mm2 of heatsink copper. The 12 thermal vias, the three 1850 mm2 planes, U1's position near board centre - every one of them moves this result by exactly **0.00 C**. What "54.17 vs 55" measures is the distance between two *declared inputs*: `power_w = 1.06` and `dt_c = 55`.

`check_thermal` is also **not additive**: U1's 54.17 C and L1's 42.51 C are each computed as if that part were the only heat source on the board. They cannot both be "the rise".

Believe the P1/P2 board-level ladder. I re-derived its inputs from scratch before looking at its answer:

```
h_conv = 1.42*(dT/L)^0.25 = 1.42*(37/0.05)^0.25 = 7.41 W/m2K
h_rad = 0.9*sigma*(358^2+323^2)*(358+323) = 8.08 W/m2K (they used 8.2)
h_tot = 15.5 W/m2K over A_eff 3.4e-3 m2 -> R_ba = 19.0 C/W
```

That reproduces their `R_ba = 19 C/W` to about 2%, from textbook flat-plate correlations, and the ladder `Tj = Ta + P_board*R_ba + dT_local + P_IC*(theta_JC,bot + R_via)` is the correct

physics for a near-isothermal board (spreading length 32 mm > the 20-25 mm board half-dimension). Worst **Tj = 97.9 C at 7 V**.

Two corrections to that ladder that nobody has applied:

1. **It still carries the 16-via array at 2.16 K/W.** The board has 12. `R_1via = 29.4 K/W` -> 12 in parallel at the same 0.85 crowding factor is ~2.88 K/W, so `+1.058 * (2.88 - 2.16) = +0.76 C`. **Worst Tj ~= 98.7 C, margin 6.3 C, not 7.1 C.** The P2 digest table was not updated for the P6 16 -> 12 change. 2. **The honest band.** `R_ba` is stated at +/-30%, and only the `P_board*R_ba` term scales: `50 + 34.7*0.7 + 6 + 7.2 = 87.5 C` to `50 + 34.7*1.3 + 6 + 7.2 = 108.4 C`. The upper edge crosses the 105 C *design* limit but stays **17 C below the 125 C conservative spec floor **and** 51 C below AP64350's actual 150 C Tj max**.

What to say to the human. The thermal risk on this board is dominated by **mounting, not layout**. 53% of the cooling path is radiation from two faces. Boxing this board, conformal-coating it, or bolting it flat against a surface removes a large share of that and costs tens of degrees - far more than any 0.83 C. That belongs in the assembly note in bold. The layout side is already 3x over its copper criteria and cannot be improved meaningfully.

L1's 42.51 C / 45 C is conservative for the same structural reason: `a_eff` counts only the 153.23 mm2 of `+5V` copper, excluding the 30.44 mm2 `/SW` pad and the 1838.91 mm2 of GND under the part. Counting `/SW` too gives 39.8 C. The number that matters for L1 is the board-level `T_L1 surface 100.7-106.5 C` against the part's 155 C rating.

—

RECOMPUTED ELECTRICALS (the brief's closing checklist)

```
| quantity | recomputed | spec | margin |
|---|---|---|---|
| inductor ripple, Vin 18 V | `5(18-5)/(18*6.8u*500k)` = **1.062 A** p-p | - | -
|
| inductor peak | 3 + 0.531 = **3.531 A** | Isat 14 A (12.3 A max-col) |
**3.5x**; constraints declare 3.6 A, conservative |
| Cin RMS | `3*sqrt(D(1-D))` maximises at D=0.5 -> Vin 10 V (inside range) =
**1.500 A** | C5-C8 share 375 mA each, ~0.7 mW each at 5 mOhm ESR | large |
| output ripple, Vin 18 V | `1.062/(8*97u*500k)` = 2.74 mV + ESR 1.062*~1 mOhm =
1.1 mV -> **~3.8 mV** p-p | 50 mV | **13x** |
| setpoint | `0.8*(1+105/20.0)` = **5.0000 V** nominal; corners with VFB 792/808
mV and 0.1% parts = **4.942 / 5.058 V (+/-1.16%)** | 4.90-5.10 V | see below |
| Tj worst | **~98.7 C** (ladder, 12-via corrected); band 87.5-108.4 C | 105 C
design / 125 C spec floor / 150 C actual | 6.3 C to design |
```

Setpoint stack - one term is missing from the record. `root.py` quotes the VFB tolerance and the resistor tolerance but not resistor **tempco**. ERA3AEB1053V and RT0603BRD0720KL are both +/-25 ppm/K; worst-case differential drift over the 55 K rise from 50 C to the ~105 C board is `2*25e-6*55 = 0.275%` = **14 mV**. Adding that plus load regulation and the uncompensated C12 -> J2 IR drop (`check_irdrop` worst `+5V` drop 8.96 mV) to the +/-58 mV corner gives **~82 mV worst case against the 100 mV window** - still passing, with ~18 mV left. Worth adding the term to the record.

SPICE: I agree it can be skipped. SIM-1 (the setpoint sweep) is a four-term worst-case sum, closed above in closed form. A sweep would add no information. The bench item that actually matters is the loop, and no SPICE model of this board closes that either - it needs a network analyser.

—

WARNINGS - full list with verdicts

Every `verify_all` warning gets a verdict here.

#	finding	verdict
W1	**‘/SW‘ pour is 10.490 mm long vs the 8 mm ceiling** (bbox 10.490 x 4.421). Excess is entirely the C1(BST)/TP2/R9 stub, 5.32 mm to -x of the SW pin; the load-carrying run U1.8 -> L1.1 is 3.73 mm. Physics benign: ‘C = e0*er*A/d = 4.85 pF‘, ‘I = C*dV/dt = 17.5 mA‘ at 18 V / 5 ns, returning into the plane 0.244 mm below. **Undisclosed anywhere.**	document as accepted deviation; do not move the BST cap
W2	**check_thermal’s 0.83 C is not a layout property** (clamped ‘a_eff‘, non-additive) - full analysis above	escalate to a documentation correction
W3	**L1’s 42.51/45 C measures the wrong path** (excludes ‘/SW‘ pad and all GND)	conservative, no action
W4	**The PM floor is not actually bounded.** The dV COUT-invariance argument is correct and I do not re-litigate it (‘2*pi*fc*COUT = R5*gm*VFB/(RTsense*VOUT)‘ is COUT-free). But PM is *not* invariant: root.py’s own table gives PM(corrected) 45.6-47.2 deg at COUT 75.0 uF vs 51.3-52.8 at 96.8 - 0.6 deg over the floor at the low corner. And [75.0, 96.8] uF is a **DC-bias band**: the low end IS the vendor’s DC-bias-only ratio (68.2%), so root.py 2.2’s remark that hot X7R and aging "push toward the low end" *names* those effects without stacking them. Stack X7R hot-end loss (~-10% at the board’s own stated 83-87 C) and 1000 h aging (~-5%) on 75.0 uF -> **~62.7 uF**, ~20% below the swept corner; on the sweep’s own slope (75k -> 82k = +9% fc for -1.5 deg PM) that is roughly **-3 deg -> PM ~42-44 deg, under the 45 floor**	no respin (R5/C2/C3 are three adjacent 0603s, already bench item #1) - but the disclosure must say COUT_min is DC-bias-only, so PM reads as unbounded, not 0.6 deg clear
W5	**DNP snubber loop ~12 mm.** ‘/SW‘ -> R9.1 (47.420,43.905) -> R9.2 (47.420,40.945) -> C16.1 (44.310,40.425) -> C16.2 GND (41.130,40.425). ‘L’ = mu0*h/w = 0.256 nH/mm‘ at w 1.2 / h 0.2444 -> ~3.1 nH + ~2 nH component ESL = ~5 nH; with 470 pF that self-resonates at **~104 MHz**, inside the band a SW snubber exists to damp. C16’s GND pad is 6.7 mm from U1’s EP	record; DNP, do not respin
W6	**C12’s ground leg is one via into a 4.41 mm2 orphan F.Cu island.** Four GND pads sit on orphans - C12.2, C2.2 (16.65), R1.2 (14.20), R4.2 (3.60) - each with exactly one stitching via, so nothing floats and three are small-signal. C12 is a 22 uF output bulk cap; one 0.3 mm via adds ~0.6-1 nH, comparable to its own impedance at 500 kHz. C10/C11/C13 (4.84 / 6.79 / 8.10 mm from L1.2) are on the main island so the bank still works	add a second via at ~(54.7, 54.6): one op, zero risk, clears one advisory
W7	**J1 and J2 carry ‘+‘ on opposite edges.** J1 pin1 (‘/VIN_RAW‘) upper at y 54.685, J2 pin1 (‘+5V‘) lower at y 59.765. Forced by the wire-entry requirement (both verified off-board on the iso render), so the mirror is not an accident. Silk does mark it: ‘+‘/‘-‘ at (34.12, 54.685)/(34.12, 59.625) and ‘-‘/‘+‘ at (61.22, 54.685)/(61.22, 59.765), plus ‘VIN 7-18V‘ and ‘VOUT 5V 3A‘, all clear of the connector bodies and still visible after assembly. Residual: this silk is the ONLY mitigation for delegate Q30, and getting ‘+‘ onto the same physical edge needs J2’s pin/net assignment swapped in the **schematic**	confirm as a decision at checkpoint 4; not a P8 defect
W8	**Every power net is routed at exactly its IPC-2152 minimum.** ‘check_current‘ min_track == required on all of them: 1.52 mm (‘/VIN_RAW‘, ‘/VIN‘, ‘+VIN‘), 2.055 mm (‘+5V‘), 2.31 mm (‘/SW‘). At JLC’s ~+/-0.05 mm etch that is -2.4% on ‘+5V‘ -> dT ~10.5 C instead of 10	acceptable; state that dT=10 C is nominal, not worst case
W9	**Two verification legs are hollow.** ‘check_pdn_z‘ checked **zero pairs** (skipped ‘+VIN‘ for "no plane pair detected", attempted nothing else) - a direct and expected consequence of the deliberate 4 x GND stackup, so it is not coverage. And ‘kc.py drc‘ has no silk test for **enclosed** silk: proved by shrinking TP4’s ring to r=0.30 inside its pad (0 violations) versus injecting crossing segments (14 violations)	say so; do not count either as a pass
W10	**‘constraints_drift‘.** I diffed both files leaf by leaf: the ONLY

```
differences are the 16 numbers of the four mounting-hole keepout rects - the
documented board-local -> absolute translation.  Everything else identical;
checks ran against the correct ‘kicad/‘ copy | justified waiver; annotate or
delete the ‘architecture/‘ twin |
| W11 | **‘decoupling.json‘’s ‘_note‘ is stale** - it says ‘+5V‘ is handled by
‘pdn=false‘, which ‘constraints.json‘ records as reverted.  Two files, opposite
statements, and the stale one is what feeds ‘check_pdn‘ and ‘check_decoupling‘ |
one-line documentation fix |
| W12 | **The 6.5 mm mounting-hole mask/copper keepout was never built** (no
rule areas exist).  Benign:  all four layers are GND at every hole and the
nearest non-GND copper is 5.38 mm vs an M3 washer radius of 3.25 mm - see the
section above | record as a knowingly-dropped constraint |
| - | **10 x ‘check_current insufficient_transition_vias‘** (all ‘GND‘, all "1
via, 3.30 A needs 7") | **justified.** Every one is a leaf tap:  five are
perimeter stitching vias at x/y 30.72/45.72/60.72 carrying no net current, and
four are the sole vias of the orphan islands above (of which only C12’s carries
meaningful current - see W6).  ‘GND‘ is ‘plane_fed‘, which keeps **pour necks
at ERROR** at the full 3.3 A, and there are **zero** pour-neck errors.
Advisory is the right severity |
```

—

THINGS I CHECKED AND FOUND CLEAN

- `kc.py drc --parity --all-track-errors`: 0 violations, 0 unconnected, schematic/PCB parity clean. Re-run by me at P8, not inherited from P7.
- `check_return_path` 0, `check_decoupling` 0 (five `+VIN` associations, loop 5.05-11.50 nH, C9 closest at 2.14 mm / 5.05 nH), `check_diffpair` 0, `check_creepage` 0.
- `check_irdrop`: `+VIN` 13.96 mV, `+5V` 8.96 mV, `GND` 1.15 mV. Against a +/-100 mV output window the sense-point-to-terminal error is ˜0.2%. No concern.
- No copper on any inner layer for any non-GND net (28 vias, all GND).
- B.Cu is a bare, uninterrupted 1850 mm2 GND face with only the four J1/J2 THT annular rings and the vias breaking it - exactly what the radiation budget needs. Note for the assembly doc: J1.1 (`+VIN`) and J2.1 (`+5V`) are exposed on B.Cu, so insulating standoffs are required if the board is mounted over metal.
- No fiducials. JLC’s SMT service does not require board fiducials and the connectors are DNP/hand-soldered, so this is a note for P9, not a finding.
- LCSC stock present for all 24 BOM lines. **L1 (FAUL1050-6R8MT, C5298292) at 763 pcs is the only thin line** - single-source, no like-for-like alternate that clears DCR <= 25 mOhm hot at 6.8 uH. Every other line clears its floor by >= 4x. Schedule risk for P9, not a board defect.

OPEN - not judgeable from the artifacts

- Loop phase margin at the true COUT floor (W4). Needs a network-analyser measurement; no model available to this project closes it.
- AO4407A dv/dt immunity - no Ciss/Crss/Coss published (already disclosed as P4 W4).
- Whether the enclosure preserves the two radiating faces (W2). The single largest thermal lever on this board and it lives outside the board.
- Whether a human ever reads the silk on this board. E2’s severity depends on it; I graded it as an error because `parts.json` says J1/J2 are hand-soldered on receipt.

8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 1 attempt(s); last 2026-08-09T06:54:57, failing 0 of 16. DFM findings by severity: warning: 16.

Fabrication outputs

Exported layers: F.Cu, In1.Cu, In2.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

Bill of materials

Comment	Designator	Footprint	LCSC
100nF 50V X7R	C1,C9	C0603	C14663
3.3nF 50V X7R	C2	C0603	C1613
10pF 50V C0G	C3	C0603	C106245
100uF 35V 80mOhm	C4	CAP-SMD_BD6.3-L6.5-W6.5-LS7.2-FD	C2982822
4.7uF 50V X7R	C5,C6,C7,C8	C1206	C29823
22uF 25V X7R	C10,C11,C12,C13,C14	C1210	C49118556
4.7uF 16V X7R	C15	C0805	C277499
470pF 1kV C0G	C16	C1206	C107177
GREEN	D1	LED0805-R-RD	C2297
BZT52C15 15V	D2	SOD-123_L2.7-W1.6-LS3.7-RD	C173427
5A slow-blow	F1	Fuse_1206_C1T_BelVendorLand	C3163312
SCREW_5.08_2P	J1,J2	CONN-TH_2P-P5.08_DB128L-5.08-2P	C395868
6.8uH	L1	IND-SMD_L11.5-W10.0	C5298292
AO4407A	Q1	SOIC-8_L4.9-W3.9-P1.27-LS6.0-BL	C16072
100k 1%	R1	R0603	C25803
105k 1%	R2	R0603	C16840
24.0k 1%	R3	R0603	C23352
200k 1%	R4	R0603	C25811
75k 1%	R5	R0603	C23242
105k 0.1%	R6	R0603	C491109
20.0k 0.1%	R7	R0603	C723637
2.2k 1%	R8	R0603	C4190
22R 1% 0.25W	R9	R1206	C17958
AP64350SP-13	U1	SO-8_L4.9-W3.9-P1.27-LS6.0-BL-EP	C2071691

1 CPL rotation correction(s) applied; no missing LCSC numbers. *fab/order.json not available.*

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake digest - sbuck-5v3a

- Function:  open-frame synchronous buck module.  7-18 V DC in (12 V nom) ->
5.0 V
+/-2% at 3 A continuous (15 W). No MCU, no digital interface.
- Artifacts:  ‘brief/brief.md‘, ‘requirements.md‘ (analyst, 9/9 sections,
```

check_requirements pass, 0 violations), ‘requirements-answers.md‘ (delegate).
- 33 open questions answered as DELEGATE per the brief’s DECISION POLICY. No safety unknown was guessed: no mains, no battery, no motors, nothing >30 V, no RF transmit. 3.0 A sits ON the >3 A threshold - treated as high-current for copper/connector sizing, no certification path triggered.
- Binding constraints: 50 x 40 mm HARD (binds at P5 board_init), 4x M3 3.2 mm NPTH, 50 C free-air ambient with no airflow, every part LCSC-stocked.
- Derived arithmetic handed forward so P1/P2 do not re-derive: D = 0.71/0.42/0.28
at Vin 7/12/18 V; Iin = 2.44/1.42/0.95 A; input-cap RMS current peaks at 1.50 A at Vin = 10 V (INSIDE the range, not at 12 V nominal); loss budget 2.05 W at the 88% efficiency floor.
- Riskiest item: thermal. 2.05 W by natural convection at 50 C ambient on a <=50x40 mm board with no heatsink is the binding case, and the brief forbids assuming the exposed pad alone closes it.
- Decisions recorded: 4-layer 1oz/0.5oz; 4 A slow-blow input fuse added beyond the brief’s protection list; fsw target band 400-700 kHz; Opus substituted for Fable across judgment roles (out of credits) - recorded, not silent.
- Second-riskiest: input and output connectors are physically identical, so a user swap is a real failure mode. Resolution is silkscreen only (Q30).

P1

P1 Research digest - sbuck-5v3a (detail: log/P1-findings-full.md)

- Roster: 3 scouts + power architect + reference-design, plus 1 re-scout to correct
magnetics against the architect’s budget. All fragments in research/.
- TPS54360 and MP1584EN are ASYNCHRONOUS per their own datasheets - disqualified.
- Every ferrite-wound inductor fails Isat/Itemp derated to 50 C; metal-composite only.
- First inductor sweep missed the DCR ceiling 2x by only searching <=6x6 mm. Re-search:
DCR <= 25 mOhm reachable (FAUL1050-6R8MT, 18.5 mOhm cold / 24.05 mOhm hot).
- AP64350 UVLO divider draws 7-10 mA (~120 mW, 6% of budget); LMR33630 does it at 23 uA.
- Loss budget IS the thermal budget: 19 C/W per watt, worst Tj 99.7 C vs 105 C limit.
- Exposed pad alone = 138 C/W -> Tj 188 C. Package rule: pad + thetaJA <= 45 C/W.
- Inrush benign (body diode forward - a gate RC cannot limit it); real hazards are the
25.4 V hot-plug ring (15 V Vgs Zener) and bulk ESR as damping (50-300 mOhm, no polymer).
- Board surface 83-87 C forces X7R everywhere; X5R is a latent failure.
- 4 research conflicts resolved explicitly (DCR, bulk ESR, dielectric, Vgs clamp).
- Left for P2: IC choice (4.0 A floor is revisitable - it was a delegate tightening),
fsw coupled to inductor stock, and vendor 2 oz layouts vs "2 oz does not help".

P2

P2 Architecture digest - sbuck-5v3a

- IC AP64350SP-13 at 500 kHz (RT=200k). Decided on MAX Rds(on) 75/45 mOhm vs LMR33630 TYPICAL 95/66; TI max column gives Tj ~112 C, over the 105 C limit. SY8205 rejected on a Preliminary datasheet with typ-only Rds.

```
- Stackup JLC04161H-1080B, 4 layer, 1 oz outer / 0.5 oz inner.  F.Cu + In1 +
In2 +
B.Cu ALL GND - the single most load-bearing entry in constraints.json.
- 1 oz beats 2 oz here:  JLC's only 4L/1.6mm 2oz-outer lamination has a 0.4284
mm
L1-L2 prepreg vs 1080B 0.2444 mm, nearly doubling thermal-via resistance.  Both
vendor reference layouts (2 oz) lose on the real stackup.
- Budget 1.646 W at 12 V = 90.1% (spec >88%).  Worst Tj 97.9 C at 7 V vs 105 C,
7.1 C margin.  Every extra 100 mW anywhere costs 1.9 C.
- ONE flat root sheet, 33 placed parts - hierarchy rejected to eliminate the
/<sheet>/<LABEL> net-name mismatch class.
- Overrode delegate answer Q14:  UVLO 6.5/6.0 V retargeted to 6.2/5.3 V. A 0.5
V
hysteresis is LESS than this board own 0.49 V cable drop at 2.44 A
(motorboating), and VON at the max corner lands ~7.0 V - at the minimum rated
input.  Side benefit:  UVLO divider draw falls from 81-181 mW to 1-2 mW.
- Corrected three P1 numbers rather than inheriting them:  thermal-via
arithmetic
had used a retired phantom stackup dielectric; spreading length 33 -> 32 mm;
U1 power 1.001 -> 1.058 W. Re-run closes 64 mW BETTER than P1.
- BOM ~6.35 USD/board.  Flagged:  delivered cost at qty 5 is 13-16 USD/board
once
PCBA setup and Extended feeder fees amortise - NRE, not parts.
- Riskiest decision:  the IC. Riskiest TASK: the compensation (vendor
Rcomp/Ccomp
are quoted for 2x22uF, this board has 5x22uF).
```

P3

```
# P3 Parts + Library digest - sbuck-5v3a

- 24 distinct parts -> 33 refdes.  11 Basic / 13 Extended.  $6.34/board at qty
1 breaks.
- fp_verify 3 passed / 0 failed / 2 pad_size warnings.  59/59 pins retyped, all
13 footprints carry courtyards, lib tables registered under ${KIPRJMOD}.
- U1 theta_JC confirmed EXACTLY 5.0 C/W (p.4 table) - the architect's ceiling
was
<=5, so the junction ladder holds with zero margin.  RT=200k -> 500 kHz
confirmed
from the datasheet's own formula.  HS current limit 4.25/5.0/5.75 A.
- TWO library defects found and fixed, both approved as hand-edits
(lib/EDITS.md):
U1 EP was pulled 12.6% undersized (3.200x2.500 vs vendor 3.502x2.613 mm) on a
zero-margin thermal design; and KiCad's stock 1206 fuse land differs materially
from Bel's own (30% narrower in the pitch axis) on a 2.44 A part.
- F1's EasyEDA record genuinely 404s (probed - a real not-found, not the 403
rate
limit).  Filled from KiCad stock + vendor land rather than re-sourcing, because
C3163312 is the only listing for a fuse confirmed ACTUALLY slow-blow.
- R6 116k -> 115k:  116k is not E96 and has zero 0603 stock; 115k/22.1k is what
the
AP64350 datasheet's own 5 V reference design specifies.
- Recorded as unknowns rather than guesses:  FB bias current (not stated
anywhere in
the datasheet), minimum off-time (not stated), thermal-via count (vendor is
only
qualitative - our 16x0.3mm array is a derivation).  Vendor thermal data is on 2
oz
copper against our 1 oz build -> deferred to P8 check_thermal on real geometry.
```

```
- Single-source risks:  L1 (763 pcs, no true second source), U1 (no
pin-compatible
alternate - differs from LMR33630 on 4 of 8 pins).
- Carried to P4:  C4 polarity must come from the footprint silk mark, not its
generic 1/2 symbol pins.
```

P4

```
# P4 Schematic digest - sbuck-5v3a

- Gates:  ‘erc‘ **0 errors / 0 warnings** (2 attempts, both pass; 2nd after the
snubber
rewire).  ‘netlist_audit‘ exit 0, 0 violations.  15 nets, 44 components, 86/86
pins
connected.  Regen identity confirmed (two builds electrically identical).
- Fresh-context review:  **0 errors / 7 warnings**.  No netlist change was
required by
any finding.  2 warnings FIXED, 5 waived -> ‘reports/erc-waivers.md‘.
- Vout = 5.0000 V exactly (105k/20.0k, both 0.1%), corners 4.9417/5.0585 V =
+/-42 mV
margin.  The reviewer showed 1% parts MISS by 32-36 mV, so 0.1% is proven, not
asserted.
- UVLO VON 6.230 / VOFF 5.325 V, 0.905 V hysteresis, verified against the
vendor’s
pull-up-inclusive Eq.2/3 - and the vendor’s own constants shown
self-consistent.
- Compensation R5=75k, C2=3.3nF, C3=10pF. The reviewer reproduced P4’s table TO
THE
DIGIT independently, then falsified its calibration:  the vendor’s Fig.29 plot
has
C4=33pF fitted, so the claimed validation compared the wrong variant.  Claim
withdrawn.
Like-for-like factor is 0.743, not 0.83 - the model is optimistic on all three
axes.
- **Honest standing of the loop**:  PM 45.6-52.8 deg vs a 45 deg floor.  Load
step
**200 mV against a 200 mV limit** on the corrected reading (148 mV modelled).
No R5
value clears fc, PM and dV together (sweep in root.py s2.5a), and dV is
INVARIANT in
COUT, so more capacitance would be pure cost.  Accepted as an at-limit
disclosure and
logged as the board’s #1 bring-up bench item; R5/C2/C3 are 3 adjacent 0603
parts.
- The fc "24.4 kHz vs 25 kHz" shortfall is NOT a requirement miss - 25 kHz was
the
orchestrator’s own proxy; the real spec is 100 us recovery and settling is 32.6
us (3.1x).
- Fixed, not waived:  DNP snubber overstressed its own resistor (162 mW at 18 V
into an
0603 rated 0.1 W). R9 -> 1206 0.25 W, C16 -> 470pF C0G 1206.  Now 76 mW, 30%
of rating.
- C3 refitted as the Eq.19 COMP pole; the reviewer measured the Eq.20
feedforward as
CONDITIONALLY UNSTABLE, not merely low-margin - a stronger result than P4 knew.
- Caveat on evidence quality:  the clean ERC is WEAK proof for U1, since nearly
every
U1 pin is ‘passive‘ in the pulled library.  Pin-level confidence comes from the
datasheet extract + the reviewer’s independent netlist read, not from the gate.
```

```
- Carried forward:  silkscreen connector labelling (Q30's ONLY mitigation for a
swapped
input/output, unreachable from a generator) -> P6/P8.  DNP filtering -> P9
manual.
SIM-1/SIM-2 -> P8.
```

P5

```
# P5 Board Setup digest - sbuck-5v3a

- `board_init`:  50 x 40 mm exactly (bbox 22.72,37.225 -> 72.72,77.225), 4
layer,
stackup JLC04161H-1080B, 1 oz outer / 0.5 oz inner, 44 components, 15 nets.
- **parity = 0** after two fixes.  Both were real:
(a) 11 x "Exclude from BOM settings differ" on H1-H4 + TP1-TP7 - the footprints
declare exclude_from_bom, the symbols did not.  Fixed at the generator
(`in_bom = False`), not by loosening anything.
(b) 2 x duplicate/value mismatch on H1 - caused by MY passing `--mounting-holes
4`
when the schematic already carries H1-H4.  Re-run with `--mounting-holes 0`.
The 4x M3 now come from the schematic as real parts; P6 must place them at
the corners, inset 3.5 mm.
- Fab floors at ERROR severity from profile `4layer_1oz`:  clearance 0.1016,
track
0.1016, via dia 0.45, hole 0.2, annular 0.1, hole-to-hole 0.5, edge 0.3 mm.
Stated explicitly because "DRC 0/0" is meaningless if the floors sit under the
fab's.
- `rules_gen`:  14 rules, capability_class 4layer_1oz, copper derived from the
stackup.
Netclasses split PER REQUIRED WIDTH (not flattened to one max-width Power
class):
Pwr_1p52mm (/VIN_RAW, /VIN, +VIN @ 2.6 A), Pwr_2p055mm (+5V, GND @ 3.3 A),
Pwr_2p31mm (/SW @ 3.6 A). Matches the architect's IPC-2152 numbers exactly.
- `diff_pairs` is an explicit empty list and stayed empty - that is what stops
/SW,
the highest-current node, from silently acquiring differential-pair gap rules
and
an inner-layer track ban.  `hv_rules` empty (no pair reaches the 30 V IPC-2221
trigger).
- Residual, NOT resolved at P5:  22 copper_edge_clearance errors + 2
silk_over_copper
warnings, all from the naive shelf pack at the outline origin corner, plus 71
unconnected.  P6 placement and P7 routing own these respectively.
- Sidecars (constraints.json, decoupling.json) sit beside the board as
required.
```

P6

```
# P6 Placement digest - sbuck-5v3a

Gate `place`:  **PASS, 0 violations.** Final DRC: 2 warnings (a footprint
defect,
not placement) + 71 unconnected (P7's).  8 of 8 edit iterations used.

## The four numbers the brief demanded

| Deliverable | Result | Limit |
|---|---|---|
| Input hot-loop enclosed area | **2.57 mm^2** (C9 out 2.14 mm, return 3.69 mm)
| minimise |
```

```
| /SW copper area | **~26 mm^2** (17.4 pour + 8.6 taps) | <= 40 mm^2 |
| /FB run | **2.21 mm** R6->U1.5; 7.04 mm incl.  R7 | short |
| Thermal vias under U1 | **12** (not 16 - see below) | - |

C9 (100 nF) sits 0.7 mm off U1's VIN pad on F.Cu, GND pad aligned to the BST|VIN
inter-pad slot, return through a 0.67 mm slot - **no vias in the loop**.  C5-C8
form
the next ring 4-6 mm out.  C4 is 22.2 mm away as constraints require.  /SW is
5.97 mm
long, >= 2.5 mm wide, 6.25 mm from the nearest edge and 9.7 mm from J2.  FB
senses at
**C12's +5V terminal**, not the inductor pin.

## Findings that changed other phases

- **16 thermal vias is geometrically impossible.** EP 3.502 x 2.613 mm; four
0.55 mm
lands at 1.0 mm pitch need 3.55 mm, and JLC's 0.5 mm hole-to-hole floor caps it
at
3x4 = 12 at 0.9-0.95 mm pitch.  R_via 2.48 vs 1.90 K/W = **+0.5 C of Tj**,
inside
the margin.  `check_thermal` would not have caught this.
- **`placement.fixed` silently disabled all 8 separation constraints** - fixed
refs
are excluded from the annealer's cluster list, so every pair landed in
`separation_unknown_refs` at zero cost weight.  Verified all 8 by hand; all
pass.
- **P5 omitted the mandatory keepout translation** - rects were still
board-local,
three landed off-board, the fourth produced a phantom violation.  Corrected.
- **Both connector rotations in `placement.edges` were wrong** (J1 0, J2 180
pointed
both wire entries INTO the board).  Correct:  J1=270, J2=90.  Proven by
orthographic
side render - the mouth is the SHORTER plastic face here, so the WRL bbox alone
would have misled.

## Judgment calls

- **Anneal rejected on merit.** Candidates hit HPWL 224.6 vs the hand
structure's
234.0, but the 3.1% is almost entirely plane-fed GND, all four carried 5-7
courtyard overlaps after repair, and cand1 pulled R7 3.5 mm closer to L1 and
lengthened /FB.
- Silk complete and DRC-verified, 112 warnings -> 2:  "VIN 7-18V" / "VOUT 5V
3A" with
polarity at both terminals, pin-1/polarity on C4/D1/D2/U1, all 7 test points
labelled by signal, board name + REV A + date.  This is Q30's ONLY mitigation
for a
swapped connector, so it was a requirement, not decoration.

## Handed to P7

- **Pour the four GND planes AND pour /SW as a zone BEFORE any autoroute.** The
0.83
route probe is an artifact:  GND is 25 of 71 connections and plane-fed, so
unpoured
Freerouting tried it as 2.055 mm tracks.  And `aiee_pwr_width_SW` (2.31 mm)
applies
to EVERY /SW track including the 0603 bootstrap cap's - a zone is not a track.
```

```
- 25 structural footprints are LOCKED; P8 fixers must unlock before nudging.
- 2 residual DRC warnings are a D2 SOD-123 footprint defect (silk clips pad 2's
mask
by ~0.05 mm) - library fix approved and in progress.
- Cosmetic:  C9's refdes label sits over C7's body; `silk_place` found no
better spot.
```

P7

```
# P7 Routing digest - sbuck-5v3a

Gate `drc_routed`:  **PASS, exit 0** - 0 errors, 0 warnings, 0 unrouted, parity
clean,
`--all-track-errors` on.  Completion **1.00** (71/71 connections closed).
Board:  115 tracks (all F.Cu), 31 vias (all GND), 14 zones, 0 unfilled.

## The numbers the brief demanded

| Deliverable | Result | Limit |
|---|---|---|
| /SW copper, F.Cu | **30.44 mm^2** (22.94 main node + 7.51 tap arm) | <= 40
mm^2 |
| /SW min cross-section, U1.8 -> L1.1 | **3.50 mm** | >= 2.5 mm |
| /SW main-node length U1.8 -> L1.1 | **5.97 mm** (bbox incl.  BST/TP2/snubber
arm 10.49) | <= 8 mm |
| /SW on inner or bottom copper | **0.000 mm^2** on In1/In2/B.Cu | zero |
| Thermal vias in U1 exposed pad | **12** (3 x 4 at 0.90 mm pitch, 0.6/0.3) | 12
|
| Stitch vias within 6 mm of U1 EP | **11** | >= 8 |
| F.Cu GND island contiguous with U1 EP | **1395.2 mm^2** | >= 100 mm^2 |
| In1 GND within 12 mm of U1 | **444.9 mm^2, ONE piece** (99.0% of the disc) |
>= 400, no split |
| B.Cu GND total | **1850.1 mm^2** | >= 1500 mm^2 |
| GND within 14.3 mm of U1, summed over layers | **2169.7 mm^2** | 529 pass /
650 A_sat |
| Vias inside the C9 -> U1 hot loop | **0** | 0 |
| GND tracks anywhere | **0** - all return is plane/pour | 0 |

Min via hole-to-hole 0.600 mm (JLC floor 0.500).  `/FB` 13.34 mm total, 3.13 mm
from
/SW copper, 3.52 mm from L1, still sensing at C12's +5V terminal.  In2.Cu came
out
**GND**, not +5V - checked explicitly.  No placement was touched; all 25 locks
intact.

## Chain that ran

`planes_gen` (7 zones) -> 12 in-pad thermal vias (`route_edit`) -> 9
hand-placed U1
ring vias -> `stitch_vias` (6 area vias, 15 mm auto pitch) -> `route_auto`
(`--power-layers In1.Cu,In2.Cu,B.Cu` so every track lands on F.Cu; FR rung 1,
124
tracks, KRT finish rejected on merit) -> pour fan-in pass B + 9 rips -> 4
island-bond
vias -> `plane_repair` (no repair needed:  0 split, 0 dead islands) -> gate.
`route_cleanup` **skipped** - DRC was already 0/0 and its loop-breaker has
regressed
twice on pour boards; nothing to gain, a real regression risk.

## Three findings (all in LEARNINGS)
```

```
- **‘planes_gen‘ rejects ‘_note‘.** ‘_PLANE_KEYS‘ is a strict whitelist, so
‘constraints.planes[]‘ - which carries this file’s most load-bearing note -
cannot be
fed to the documented step-0 invocation at all.  Used planes-only sidecars
(‘kicad/planes_p7.json‘, ‘kicad/planes_p7b.json‘) instead of stripping the
notes.
- **A clean ‘--pad-window‘ does not predict Freerouting’s width.** All 61 power
pads
probed ‘ok‘ (R6.1/R8.1 reported 6.97/8.00 mm against a 2.055 mm floor); FR then
routed
those 0402 taps at **0.8058 mm - the pad width** - and necked +5V to 1.6348
(1210 pad)
and +VIN to 1.4848 (0805 pad).  9 ‘track_width‘ errors, fixed by pour fan-in
per the
remediation ladder step 4, not by necking or by weakening a rule.
- **‘planes_gen‘’s via grid can only be odd x odd.** It is centroid-centred and
pitch-stepped, so U1’s 2.613 x 3.502 mm pad yields 3 x 3 = 9, never the 12 that
fits.
Hand-placed the 3 x 4 array with a half-pitch y stagger.

## Judgment calls

- **Power is poured, not trunked.** +VIN (2 rects), +5V (3 rects) and /SW (2
rects) are
F.Cu zones at priority above the GND pours.  This is the only way to satisfy
net-wide
‘aiee_pwr_width_*‘ floors at 0402/0603 stubs without touching the ‘.kicad_dru‘.
The
‘.kicad_dru‘, ‘.kicad_pro‘ netclasses and ‘constraints.json‘ are byte-unchanged.
- **All four GND planes kept solid; every track on F.Cu.** ‘--power-layers‘ was
forced to
include B.Cu (the ‘auto‘ heuristic only ever considers INNER layers, so B.Cu
would have
been fair game for Freerouting and the second radiating face would have been
cut).
- **F.Cu GND uses thermal relief except where it must not.** Board pour is
priority 0
thermal (J1/J2 screw terminals stay hand-solderable); a priority-3 ‘connect:
solid‘
island covers U1’s EP, PGND pin and C9’s return, and two priority-4 solid
patches clear
the ‘starved_thermal‘ on R1.2 / C3.2.
- **M3 washer keepouts not added as rule areas.** All four planes are GND, so a
screw or
washer bridging pour to pour is a GND-to-GND short.  Verified instead that the
nearest
non-GND copper to each hole is 5.38 / 6.89 / 10.18 / 11.60 mm - all well past
the
3.25 mm washer radius.  Flagged for P8 to confirm the reasoning rather than the
rule.

## Handed to P8

- 2 ‘/SW‘ tracks at exactly 2.31 mm survive inside the pour (FR’s U1.8 -> L1.1
legs).
Legal and load-bearing; they are counted in the 30.44 mm^2.
- The ‘/SW‘ bbox is 10.49 mm end to end because P6 put the BST cap, TP2 and the
DNP
snubber on the west side of the node.  The switching current path is the 5.97
```

mm leg;
the arm is 7.51 mm^2 of quiet tap copper. Judge the ceiling on area, which
holds.
- ‘+5V‘, ‘+VIN‘ fan-in pours mean check_current will see pour necks, not
tracks;
‘pour_neck‘ only measures between via attachments and these pours are via-free.

P8

P8 Verification digest - sbuck-5v3a

Gates: ‘verify‘ PASS (0 failing, 2 waivers), ‘drc_routed‘ PASS 0/0. 2 of 3
fix-loop
attempts used. Fresh-context review: 2 errors / 12 warnings, both errors
FIXED.

The review found two errors that every gate had passed

- **CRITICAL - D2’s ‘K‘ cathode marker was on the ANODE end.** Silk said pad 2
(‘/QGATE‘); the cathode is pad 1 (‘+VIN‘). The netlist was always right.
Fitted per
that silk, the 15 V Zener reverses to a forward diode holding Vgs at -0.7 V, Q1
never enhances, 2.6 A runs in its body diode at ~2.1 W in an SO-8 - Q1 and the
reverse-polarity protection both destroyed. Moved to (45.900, 59.000): 0.640
mm from
pad 1, 3.036 mm from pad 2, agreeing with the footprint band and pin-1 dot.
- **8 refdes labels named the wrong part** (‘silk_misattributed‘ WARNINGS, so
the
error-only gate passed them). 8 fixed, 3 residual - at 1.0 mm text the label
is
2.6-3.2 mm wide against a 2.5-3.0 mm passive pitch, so no clean position
exists.

Machine findings, and what they turned out to be

- ‘check_thermal‘ L1: REAL. 100.7 mm^2 of +5V copper -> 48.10 C rise vs 45
allowed.
Grew a +5V pour around L1’s body -> 153.3 mm^2, **42.51 C**.
- ‘check_current‘ GND x2: a MEASUREMENT ARTIFACT over three real defects.
‘pour_neck‘
erodes each zone fill in isolation, so a pour split across zone objects reports
a
phantom 0.00 mm neck. Underneath were three genuine sub-2.055 mm via
attachments in
1.06-1.58 mm fingers. Repartitioned the zones; 3 unhomeable ring vias deleted.
- ‘check_silk‘ x7: CHECKER DEFECT (check_silk.py:172 buffers an unfilled circle
into a
filled disc). Waived, then the underlying rings were deleted at P9 anyway.
- ‘check_pdn‘ x2: waived. Corrected reason - it never looks at the board, it
filters
decoupling.json and errors on an empty list, so it is a TRUE statement that no
association named those rails. Coverage comes from check_irdrop instead.

IR drop on the real copper (all three power nets)

Net	I	R	worst drop	jmax
+VIN	2.60 A	5.36 mOhm	14.0 mV	2.65 A/mm
+5V	3.30 A	2.71 mOhm	**9.0 mV**	2.41 A/mm
GND	3.30 A	0.34 mOhm	**1.1 mV**	0.46 A/mm

```
‘check_pdn_z‘ passes.  Richardson deltas 0.03-0.05, so the grids converged.

## Self-correction worth recording

Setting ‘pdn:  false‘ on +5V and GND silenced check_pdn’s finding but ALSO
excluded
both nets from check_irdrop - the two highest-current nets on the board.
Reverted and
waived instead:  a waiver says "this check does not apply" without disabling
unrelated
analysis.

## Thermal, restated honestly

check_thermal’s "0.83 C headroom" on U1 is NOT a layout property:  a_eff
measures
2137 mm^2 and clamps to A_SAT 645, so theta is constant for any 4L board above
645 mm^2 - the 12 vias and three planes move it by 0.00 C. The believable
figure is
the P1/P2 board ladder, corrected for 12 vias rather than 16 (2.16 -> 2.88
K/W):
**Tj ~98.7 C, margin 6.3 C** against the 105 C design limit, with a +/-30% band
of
87.5-108.4 C. Spec floor is 125 C and abs max 150 C. The dominant lever is
MOUNTING,
not layout - 53% of cooling is radiation from two faces.
```

P9

```
# P9 DFM digest - sbuck-5v3a

Gate ‘dfm‘:  **PASS, 0 failing / 16 total** (16 warnings), measured on gerbers
re-exported after the fixes.  1 of 3 fix-loop attempts used.

## Two real findings, both invisible until the gerbers existed

**1.  Silk over the solder-mask opening, 7 test points, 0.1798 mm^2 each.**
My diagnosis was WRONG and the fixer corrected it.  I assumed an oversized mask
aperture (~0.92 mm radius).  Measured:  the aperture is **1.7664 mm^2 = exactly
pi*0.75^2, zero mask expansion**.  The real cause is
‘lib/gerblib.py:read_gerber‘
approximating drawn arcs **by their chord** - KiCad exports a full circle as
two
180-degree arcs, so both chords collapse onto the DIAMETER, a phantom 1.5 x
0.12 mm
bar through the pad centre.  0.1798 = 1.5 x 0.12 exactly.
My proposed fallback (enlarge the ring) **could never have worked** - the chord
runs
through the centre at any radius.  Deletion of the 7 decorative rings was the
only
board-side fix.  The signal-name labels, which carry the actual information,
are untouched.

This is a SECOND false positive on the same innocent ring, from a DIFFERENT
tool
defect than the P8 ‘check_silk‘ one.  **Two checkers agreeing was not
corroboration.**

**2.  Copper clearance 0.0127 mm at (39.67, 51.05).**
```

```
Both features were GND. The second was a **0.0029 mm^2 crumb of the priority-0
board
pour**, trapped in the 0.02 mm lane between zone 2's top (51.1) and the P8 Z2's
bottom
(51.12).  Mechanism:  **KiCad deburrs a pour at min_thickness BEFORE
subtracting
higher-priority zone outlines**, so a sub-min_thickness lane between two zone
outlines
gets refilled and is never deburred away.  The P8 "clears zone 2 by 0.02 mm"
choice is
exactly what manufactured it.  Fixed with a priority-6 GND bridge zone
straddling the
seam; a second, longer 0.02 x 1.93 mm hair that nothing had flagged went with
it.
F.Cu GND islands 5 -> 4.  Min island gap 0.01267 -> **0.1021 mm**.

## Fab package

`fab/`:  gerbers + zip, drill, `BOM.csv` (21 rows), `CPL.csv` (29 rows),
`BOM-all.csv` / `CPL-all.csv` (unfiltered, for audit), `DNP-NOT-ASSEMBLED.txt`.
One rotation correction applied against JLCPCB's convention:  **Q1 SOIC-8, +270
deg**.
`bom_complete:  true`, no missing LCSC codes.

**DNP filtering was done BY HAND** - no script in the pipeline reads a DNP
mark, so
left alone JLC would have populated both hand-solder screw terminals and the
snubber.
Excluded:  J1, J2 (THT, hand-soldered on receipt per Q25), R9, C16 (snubber,
populate
only if bring-up shows SW ringing).

## Disclosed, deliberately NOT fixed

- Two residual etch slivers (0.05 x 1.71 mm and 0.078 x 0.497 mm), fused to
neighbouring copper so no gate fires, but a fab may query them.
- **Min F.Cu island gap is 0.1021 mm against JLC's 0.1016 floor - 0.5 um of
margin.**
Pre-existing and unchanged by this work, but it means ANY further pour edit
risks
tripping `dfm_clearance`.  The risk of touching exceeds the risk of leaving.
- 16 dfm warnings remain (15 pin-1 dot slivers + 1 silk width), unchanged.

## Script defects found, recorded not fixed (skill work belongs in its own
session)

- `lib/gerblib.py` chords drawn arcs - the S12 `approximate_arcs` fix reached
`_flash_polys` only.  **Every footprint drawing a silk ring around a pad will
fail
the dfm gate forever** until this is fixed in the tool.
- `planes_gen` should refuse or warn when a planned region edge lands within
`min_thickness` of an existing same-layer zone outline - fully detectable at
plan time.
- No script owns footprint-internal silk on a placed board.
```

P10 *(no digest recorded)*

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-09T01:31:22	erc	pass	0/0
2026-08-09T02:14:15	erc	pass	0/0
2026-08-09T03:22:36	place	pass	0/0
2026-08-09T04:07:57	drc_routed	pass	0/0
2026-08-09T04:09:23	drc_routed	pass	0/0
2026-08-09T04:58:04	drc_routed	pass	0/0
2026-08-09T04:58:04	verify	pass	0/26
2026-08-09T04:59:41	verify	pass	0/28
2026-08-09T06:30:36	drc_routed	pass	0/0
2026-08-09T06:30:36	verify	pass	0/23
2026-08-09T06:54:57	dfm	pass	0/16
2026-08-09T06:54:58	verify	pass	0/16

Issues

Id	Phase	Kinds	Severity	Status
1	P8	pour_neckdown	error	fixed
2	P8	thermal_area	error	fixed

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-08T23:30:37	DELEGATE approval under the brief's DECISION POLICY (no clarifying questions). Architecture, stackup, IC and constraint set accepted as written. User may still interject - nothing is fabricated yet.
2	P4	approved	2026-08-09T02:14:45	DELEGATE approval per the brief's DECISION POLICY. erc 0/0, review 0 errors/7 warnings, 2 fixed + 5 waived. Load step accepted AT the 200mV limit on the corrected reading - disclosed, not smoothed.
3	P6	not reached		
4	P8	approved	2026-08-09T06:58:09	DELEGATE approval per the brief's DECISION POLICY. All gates pass. Load step AT the 200mV limit and loop PM unbounded at the true Cout floor - both disclosed, bring-up bench item #1.
5	P10	not reached		

Generation cost

Not recorded (no reports/cost.json).

10 Artifact Index

File	Size	Note
`state.json`	62,747 B	
`kicad/PCB-0008-B_sbuck-5v3a.kicad_pcb`	423,746 B	
`kicad/PCB-0008-B_sbuck-5v3a.kicad_sch`	153,179 B	

File	Size	Note
fab/PCB-0008-B_sbuck-5v3a_gerbers.zip	80,994 B	
fab/BOM.csv	790 B	
fab/CPL.csv	971 B	
brief/brief.md	5,387 B	
requirements.md	20,496 B	
architecture/blocks.md	12,147 B	
architecture/constraints.json	18,235 B	
architecture/decisions.md	15,216 B	
architecture/power_tree.md	12,708 B	
architecture/sheets.md	10,238 B	
architecture/stackup.md	7,559 B	
reports/board_init.json	64,534 B	
reports/board_update_dryrun_D2check.json	352 B	
reports/bom_cpl.json	12,320 B	
reports/erc-p4.json	263 B	
reports/erc-waivers.md	8,048 B	
reports/fab_export.json	2,546 B	
reports/fix_dispatch_verify.json	874 B	
reports/fp_verify_J1J2.json	727 B	
reports/fp_verify_Q1_jedec.json	1,406 B	
reports/fp_verify_U1.json	3,241 B	
reports/fp_verify_U1_v2.json	3,253 B	
reports/gate-dfm.json	366 B	
reports/gate-drc_routed.json	382 B	
reports/gate-erc.json	1,615 B	
reports/gate-place.json	1,367 B	
reports/gate-verify.json	1,720 B	
reports/lib_pin_types.json	5,946 B	
reports/lib_pin_types_v2.json	628 B	
reports/lib_pull_F1_retry1.json	830 B	
reports/lib_pull_remainder.json	8,572 B	
reports/lib_pull_snubber_v2.json	3,427 B	
reports/netlist_audit-p4.json	1,062 B	
reports/p6_drc_final.json	52,879 B	
reports/p6_drc_post_silk.json	51,413 B	
reports/p6_drc_pre_silk.json	130,235 B	
reports/p6_edit1.json	5,044 B	
reports/p6_edit2.json	1,265 B	
reports/p6_edit3.json	2,265 B	
reports/p6_edit4.json	2,909 B	
reports/p6_edit5.json	3,270 B	
reports/p6_edit6.json	2,443 B	
reports/p6_edit7.json	251 B	
reports/p6_edit8.json	266 B	
reports/p6_fix2.ops.json	899 B	
reports/p6_fix3.ops.json	1,683 B	
reports/p6_fix5.ops.json	1,994 B	
reports/p6_fix7.ops.json	146 B	
reports/p6_fix8.ops.json	117 B	
reports/p6_lock.ops.json	1,534 B	
reports/p6_silktext.ops.json	3,223 B	
reports/p6_struct.ops.json	3,862 B	
reports/p7_bond_vias.json	612 B	
reports/p7_bond_vias.ops.json	500 B	
reports/p7_drc_after_planes.json	30,628 B	
reports/p7_drc_b.json	3,227 B	
reports/p7_drc_c.json	393 B	
reports/p7_pad_window.json	8,228 B	

File	Size	Note
reports/p7_plane_repair.json	3,310 B	
reports/p7_planes_gen.json	1,836 B	
reports/p7_planes_gen_b.json	1,755 B	
reports/p7_render.json	315 B	
reports/p7_rip_necks.json	669 B	
reports/p7_rip_necks.ops.json	732 B	
reports/p7_route_auto.json	8,870 B	
reports/p7_stitch.json	1,482 B	
reports/p7_stitch_dryrun.json	1,470 B	
reports/p7_thermal_vias.json	1,422 B	
reports/p7_thermal_vias.ops.json	1,434 B	
reports/p7_u1_ring.json	1,117 B	
reports/p7_u1_ring.ops.json	1,056 B	
reports/p8_drc_check.json	380 B	
reports/p8_fix1_d2k.json	343 B	
reports/p8_fix1_d2k.ops.json	550 B	
reports/p8_fix1b_d2k.json	340 B	
reports/p8_fix1b_d2k.ops.json	736 B	
reports/p8_fix2_silk.json	1,303 B	
reports/p8_fix2_silk.ops.json	1,442 B	
reports/p8_fix2_silk_dryrun.json	4,669 B	
reports/p8_fix2_silk_pass2.json	349 B	
reports/p8_fix2_silk_pass2.ops.json	269 B	
reports/p8_fix3_c12via.json	289 B	
reports/p8_fix3_c12via.ops.json	565 B	
reports/p8_fix_render.json	281 B	
reports/p8_gate_drc_after.json	382 B	
reports/p8_gnd_vias.json	509 B	
reports/p8_gnd_vias.ops.json	387 B	
reports/p8_irdrop.json	4,695 B	
reports/p8_pdnz.json	619 B	
reports/p8_planes_5v.json	561 B	
reports/p8_planes_gnd.json	763 B	
reports/p9_dfm_full.json	11,133 B	
reports/p9_planes_seam.json	561 B	
reports/place_anneal.json	4,817 B	
reports/place_seed.json	11,394 B	
reports/place_seed.ops.json	2,022 B	
reports/render_final.json	761 B	
reports/review-board.json	14,861 B	
reports/review-board.md	25,420 B	
reports/review-schematic.json	5,178 B	
reports/review-schematic.md	20,610 B	
reports/route_probe.json	311 B	
reports/rules_gen.json	1,827 B	
reports/sbuck-5v3a-schematic.pdf	116,910 B	
reports/silk_place.json	5,650 B	
reports/verify-waivers.json	3,194 B	
reports/verify_all.json	3,758 B	